Learn the Local Law, Infer Globally: Exact Diffusion Scores for Markov Data by Belief Propagation

Anonymous Authors[1]

Abstract

Diffusion models are trained to approximate the score of a noised data law, a target that is unknown for almost every distribution of interest. We study data for which it is known: a first-order Markov chain with an arbitrary continuous transition kernel under coordinatewise Ornstein–Uhlenbeck noise. Conditioning leaves the posterior factor graph a tree, so one forward–backward sweep of belief propagation (BP) returns the exact score, up to a stated quadrature, and the same sweep supplies the exact gradient that lets the kernel be learned from noised sequences alone by expectation–maximisation (EM): learn the local law, infer globally. The Gaussian kernel is a sanity check, because BP with Gaussian messages computes the linear score and the Kalman smoother. We test the approach on two chains whose scores are known: a linear chain with non-Gaussian noise, which separates what diffusion noise does to correlation from what it does to shape (the Fisher information about them decays as e^{-4t} and e^{-8t}), and the one-dimensional ϕ^4 field, a physical model with domains and kinks whose lattice snapshots are exact Markov chains. On both, EM–BP with a kernel of a few parameters beats the neural denoisers it is compared with (six on the linear chain, two on the field) at every sample size; at the estimator's own information interface (one noised view per sequence, no clean chain, no true score for model selection) the best network is a factor 7.9 behind on the linear chain. The kernel family matters: the linear-drift mixture that suffices there only reaches the linear floor on the field, where the five-parameter quartic site energy reaches 0.032–0.169 of it. A neural kernel earns its parameters only where the simple family is misspecified. We report what the structure costs (BP is slower per sequence than a network, and a coarser grid cuts its time by 6.2× without loss for the field kernels) and where it stops: a latent shared by the whole sequence leaves a bias that no sample size removes, and we repair it by augmenting the state.

1. Introduction

A diffusion model generates by reversing a noising process, and the reversal is driven by the score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ of the noised data law (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021). Training regresses a network onto that score through denoising (Vincent, 2011), so every question about what the network has learned is a question about the distance between what it computes and the score it should have computed. For real data that target is unknown: a large error cannot be told from a hard target, and the solvable alternatives are Gaussian, whose scores are linear, or mixtures, whose nonlinearity lives in a discrete label rather than in any dependence between coordinates.

The structural thesis: learn the local law, infer globally. Data with a one-dimensional dependence structure sit between those cases. If the clean sequence is a first-order Markov chain with transition kernel $K(a' \mid a)$, then under coordinatewise Gaussian noising the posterior factor graph is a tree, the score is a posterior mean (Tweedie's identity), and one forward–backward sweep of belief propagation (BP) returns it exactly for any continuous kernel, including those whose score has no closed form (Section 3). A network that denoises such data must learn the composition of local transitions into a global posterior from examples. The structured estimator learns only the local law and delegates the composition to an exact algorithm. When the kernel

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

is unknown it is learned from noised sequences alone by expectation–maximisation (EM) around the same sweep (Section 5); the result, EM–BP, is the paper's estimator (Figure 1).

Two laboratories with exact scores. The framework is stated for a general kernel (Section 2); we test it on two concrete cases chosen because their scores can be computed exactly and compared with everything else. The first is a linear chain with non-Gaussian driving noise, whose covariance is fixed by two numbers and whose shape by the noise law, so Gaussian, Laplace, Student-t, uniform and bimodal noise share one covariance exactly and differ only in higher-order structure; it separates, by construction, what diffusion noise does to correlation from what it does to shape (Section 6). The second is the one-dimensional ϕ^4 field, the stochastic Allen–Cahn model, whose lattice snapshots are exactly Markov, whose kernel is nonlinear (a double well with domains and kinks), and which has a physical vocabulary (temperature, correlation length, kink density) and exact transfer-matrix references (Section 7).

Contributions.

(i) Exact denoising for any Markov kernel. Score estimation for Markov data reduces to posterior means, which BP computes exactly up to a quadrature we measure (Section 3); the Gaussian kernel is a clean sanity check, since BP with Gaussian messages is the linear score and the Kalman smoother (Propositions 1 and 2; Section 4).

(ii) The kernel is learnable from noise, and learning it works. EM through the same sweep recovers the kernel from one noised view per sequence (Section 5). With a kernel of a few parameters it beats the neural denoisers it is compared with (six on the linear chain, two on the field) at every sample size, by a paired factor 7.9 at EM–BP's own information interface on the linear chain and by 14–24 on the field (the best network over the site energy at $M = 2048$).

(iii) The kernel family is the real modelling choice. The linear-drift mixture is exactly right for the linear chain and sits on the linear floor on the field; the five-parameter site energy, the physicist's pair energy, is best on every field cell; an unrestricted or neural kernel earns its parameters only where the simple family is misspecified (Sections 7 and 9).

(iv) Cost and limits, measured. BP is slower per sequence than a network; a coarser grid recovers most of the gap for the field kernels, and a Fourier representation of the messages is not worth building (Section 8). A latent shared by a whole sequence leaves a bias that no sample size removes, which augmenting the state repairs (Section 9).

Supporting theory: what noise erases. Why structure matters most where the noise has not yet erased it is a general fact about the Ornstein–Uhlenbeck channel, and we prove it in the laboratory where it can be checked: the channel attenuates the Hermite moment of degree m by e^{-mt} exactly, so the Fisher information a noised sequence carries about a feature first visible at degree m decays as e^{-2mt} with an explicit constant (correlation as e^{-4t}, non-Gaussian shape as e^{-8t}), and the gap between the true score and that of the covariance-matched Gaussian is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ (Section 6.1). We treat this as supporting rather than central: it explains where the structured estimator gains, not why it is exact.

2. Setting: a Markov chain with a general kernel

Data. A clean sequence $\mathbf{a} \in \mathbb{R}^L$ is a first-order Markov chain with initial law μ and transition kernel K,

$$P_0(\mathbf{a}) = \mu(a_1) \prod_{i=2}^{L} K(a_i \mid a_{i-1}), \tag{1}$$

where $K(\cdot \mid a)$ is a continuous density for every a. Nothing else is assumed in Sections 3 to 5; stationarity, symmetry and reversibility are properties of the examples below, not of the method. The kernel is the whole model: it is the one function from which the law of every site, every pair of sites and the whole sequence follows. We write $\mathbb{R}$ for the state space and L for the chain length; the ambient dimension is L, and M denotes the number of observed sequences.

The channel. The forward process is the coordinate-wise Ornstein–Uhlenbeck channel

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}), \tag{2}$$

so that $\mathbf{x}_t$ is a noised copy of the same sequence at level t. P_t is its law, and the score is $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$.

First instance: a linear chain with non-Gaussian noise. The simplest kernel with a nontrivial score is a linear one with non-Gaussian driving noise,

$$K(a' \mid a) = p_\varepsilon(a' - \rho a), \qquad a_1 \sim \mathcal{N}(0, 1), \quad |\rho| < 1, \tag{3}$$

with a driving-noise density p_ε of mean zero and variance $q = 1 - \rho^2$. The variance choice makes $\mathrm{Var}(a_i) = 1$

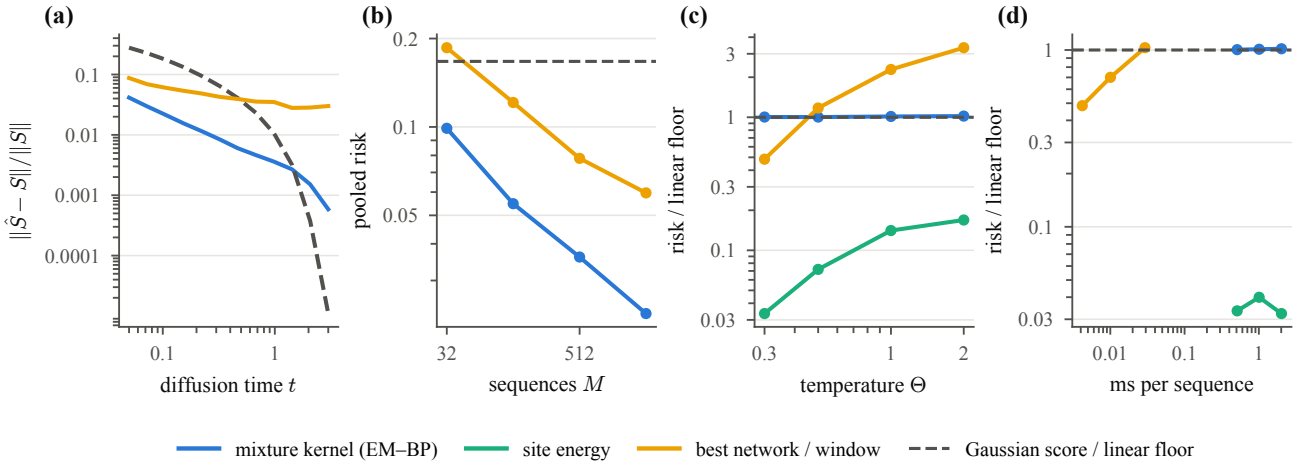


Figure 1. Summary. Relative score error against the exact score, pooled over a twelve-level noise schedule. (a) Per-level error on the linear chain with Laplace noise ($M = 2048$ training sequences): the learned kernel (blue) beats the best network (yellow) at every level and the Gaussian score (dashed) except at the highest noise levels. (b) Pooled error against the number of training sequences on the same chain. (c) The ϕ^4 field ($M = 2048$): risk divided by the linear (Gaussian) floor against temperature Θ; the five-parameter site energy (green) is far below it, the linear-drift mixture kernel of (a, b) sits on it, and the window network rises above it when the field is hot. (d) The same ratio against inference time on three lattice spacings at $\Theta = 0.3$ (eight threads, the largest network variants).

and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly for every noise law, so the families we use—Gaussian, Laplace, Student-t, uniform, a symmetric two-component mixture, and the generalised Gaussian $p_\varepsilon(\varepsilon) \propto \exp(-|\varepsilon/s_\beta|^\beta)$ whose exponent β moves the excess kurtosis from $+3$ (Laplace) through 0 (Gaussian) to -1 at $\beta = 6$, at fixed variance—share one covariance $\mathbf{\Sigma}_0$ and, every noise law being symmetric, differ only through cumulants of order four and higher. The laboratory therefore separates, by construction, what diffusion noise does to correlation from what it does to shape. Throughout Section 6 the frozen configuration is $\rho = 0.85$, $L = 32$, Laplace noise.

Second instance: a physical field. The one-dimensional ϕ^4 field is the stochastic Allen–Cahn equation $\partial_\tau \phi = \partial_s^2 \phi - V'(\phi) + \sqrt{2\Theta}\,\eta$ with the double well $V(\phi) = (\phi^2 - 1)^2/4$ at temperature Θ; its stationary law is the Gibbs measure $\propto e^{-H[\phi]/\Theta}$ with $H[\phi] = \int [\frac{1}{2}(\partial_s \phi)^2 + V(\phi)]\,\mathrm{d}s$. On a lattice of spacing h the energy couples only nearest neighbours, $H_h = \sum_i [(\phi_{i+1} - \phi_i)^2/2h + hV(\phi_i)]$, so a spatial snapshot is exactly a first-order Markov chain in the site index, with a kernel of the form Gaussian step times a function of the arrival point,

$$K(\phi' \mid \phi) = \frac{\mathcal{N}(\phi'; \phi, \Theta h)\; g(\phi')}{Z(\phi)}, \qquad Z(\phi) = \int \mathcal{N}(\phi'; \phi, \Theta h)\; g(\phi')\,\mathrm{d}\phi', \tag{4}$$

with $g = \psi_0 e^{-hV/2\Theta}$ built from the leading eigenfunction ψ_0 of the symmetric transfer operator $\mathsf{T}(\phi, \phi') = e^{-hV(\phi)/2\Theta} e^{-(\phi'-\phi)^2/2\Theta h} e^{-hV(\phi')/2\Theta}$ (Appendix L). Two knobs set the chain: the temperature Θ and the spacing h; the correlation length in sites is $\xi = 1/\log(\lambda_0/\lambda_1)$. There is no phase transition in one dimension: ξ is finite at every $\Theta > 0$ and grows roughly like $e^{E_{\mathrm{kink}}/\Theta}$ as Θ falls, a sharp crossover. Unlike Equation (3), the field's expected next value is a nonlinear function of the current one, and its driving noise is state-dependent; the two families meet only at the Gaussian chain (Appendix L). Snapshots are standardised to unit variance, $a = \phi/\sqrt{\mathrm{Var}\,\phi}$.

What is scored. For an estimator $\hat{S}$ the risk is the relative root-mean-square score error against the exact score, summed over test chains, sites and the schedule of noise levels (Appendix C). Three reference points bracket every estimator: the isotropic score $-\mathbf{x}$, which ignores the chain; the linear (LMMSE) score of the true covariance, which by Proposition 1 is what any second-order estimator can reach; and the exact score, at zero (Appendix D).

3. Exact inference: from the kernel to the score

The score is a posterior mean. Tweedie's identity gives $S(\mathbf{x}, t) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$, with $\langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$ (Robbins, 1956; Efron, 2011). Estimating the score of a Markov chain under Equation (2) is there-

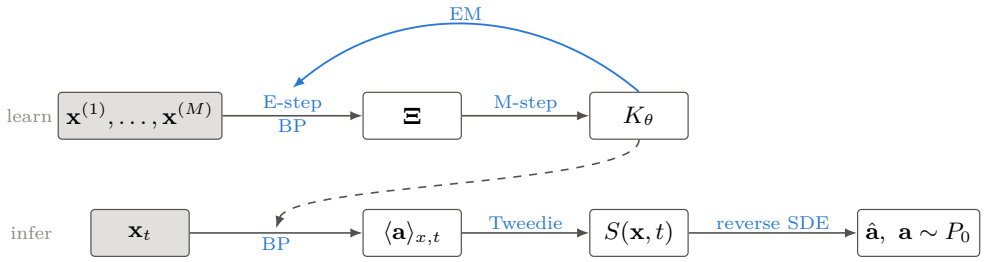


Figure 2. Learn the local law, infer globally. Top: EM on M noised sequences; the E-step runs BP and sums its pair marginals into Ξ, the M-step updates the kernel, and the two alternate. Bottom: the learned kernel inside BP at any level t gives the posterior mean, Tweedie's identity turns it into the score, and the reverse SDE uses the score to denoise or generate. A direct score network learns the whole of the bottom row at once; here only the kernel is learned.

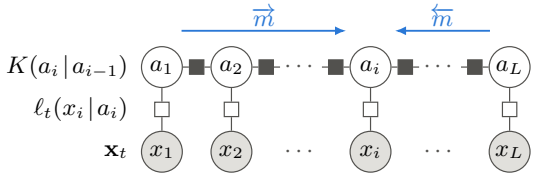


Figure 3. Why the score is exact. Latent sites (white) carry the chain prior through its transition factors (filled squares). Coordinatewise noising attaches one unary factor per site (open squares), joining a_i to its own observation x_i (shaded), and adds no edge among the latent variables. The posterior factor graph is therefore the prior's chain, a tree, so the forward and backward sweeps $\overrightarrow{m}, \overleftarrow{m}$ are exact; meeting at site i they give $\langle a_i \rangle_{x,t}$, and Tweedie's identity returns $S(\mathbf{x}, t) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$ exactly. The interior of the chain is elided; L is its length.

fore the same problem as computing a posterior mean, that is, as denoising.

The posterior is a tree. Conditioning Equation (1) on $\mathbf{x}$ multiplies the prior by one unary factor $\ell_t(x_i \mid a_i) \propto \exp[-(x_i - e^{-t}a_i)^2/2\Delta_t]$ per site (Figure 3): no edge is added among the latent variables, the posterior factor graph is the prior's chain, and sum-product on it is exact (Pearl, 1988; Kschischang et al., 2001). Two sweeps of the functional messages

$$\begin{aligned} \overrightarrow{m}_i(a) &\propto \int K(a \mid a')\, \ell_t(x_{i-1} \mid a')\, \overrightarrow{m}_{i-1}(a')\, \mathrm{d}a', \\ \overleftarrow{m}_i(a) &\propto \int K(a'' \mid a)\, \ell_t(x_{i+1} \mid a'')\, \overleftarrow{m}_{i+1}(a'')\, \mathrm{d}a'', \end{aligned} \tag{5}$$

with $\overrightarrow{m}_1 \equiv \mu$ and $\overleftarrow{m}_L \equiv 1$, return the site marginals $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$ and the pair marginals $\propto \overrightarrow{m}_i \ell_t K \ell_t \overleftarrow{m}_{i+1}$ of the whole sequence. Their means give the exact score of a correlated, possibly strongly non-Gaussian, L-dimensional law, for any kernel, even one whose score has no closed form. The pass is a smoother, not a filter: every site is conditioned on the whole sequence. It needs the kernel only through the integral operator in Equation (5), which is why everything that follows treats K as the single object to be known, parameterised or learned.

What "exact" means here. Messages are functions on $\mathbb{R}$. We carry them on a grid of N_g points on $[-A, A]$ with composite trapezoidal weights, so three statements are kept apart (Appendix B): sum-product is exact on functional messages; forward–backward is exact for the discretised factor model, whose evidence is exact to floating point; and that model approximates the continuous chain through truncation (mass beyond $\pm A$) and quadrature (second order in the spacing), both of which we measure rather than assume. At $N_\mathrm{g} = 401$ points on $[-8, 8]$, the operating grid of every experiment below, the reference is accurate to 10^{-5} relative on ordinary draws for every driving-noise family of Equation (3); on a Gaussian chain, where two independent exact answers exist, it is accurate to 1.9×10^{-13} (Section 4); and on the ϕ^4 field it agrees with a grid four times finer to 1.3×10^{-12} on every cell (Section 7). The domain, not the spacing, is what a heavy-tailed family tests, and at $\rho = 0$ the Student-t family needs a wider one (Appendix B). Throughout, "the exact score" means this tree-exact inference, evaluated by validated quadrature, and "the oracle" is BP with the true kernel. The cost is $O(LN_\mathrm{g}^2)$ per sequence and level, and one sweep also returns the pair marginals that EM needs (Section 5); Section 8 measures it.

Sampling. The same recursion samples. Drawing $a_1 \sim \mu$ and then each a_i from $K(\cdot \mid a_{i-1})$ on the grid is exact ancestral sampling from the chain, and a forward-filter backward-sample pass draws whole sequences from the posterior given $\mathbf{x}$, which gives uncertainty as well as a mean (Appendix L, on the ϕ^4 field). On a chain the reverse process is therefore needed to learn the kernel from noised data, not to sample from it once it is known (Section 6).

4. A sanity check: the Gaussian case

The covariance twin and the price of second-order closure. The Gaussian chain with the same ρ and q as Equation (3) has the closed-form score $S^{\mathrm{G}}(\mathbf{x},t) = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, $\boldsymbol{\Sigma}_t = e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}$, and posterior mean $e^{-t}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, the linear minimum-mean-square (LMMSE) estimator. It is the natural approximation of a non-Gaussian chain by something solvable, and BP shows exactly what it computes.

Proposition 1 (Gaussian closure computes the LMMSE of the covariance twin). Let Π_{G} project a density onto the Gaussian with its mean and variance, and let moment-projected sum-product apply Π_{G} to each message after every transition integral in Equation (5). For the chain Equation (3) with $\rho \neq 0$ and any driving-noise law of mean zero and variance q, it returns exactly the LMMSE estimator $e^{-t}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, the same for every such law.

The proof (Appendix A) is an induction over the two sweeps: a convolution's first two moments depend on the driving noise only through its first two moments, so every projected message equals the Gaussian chain's message. Two consequences are used as checks below. Sum-product with any Gaussian kernel is a linear map of $\mathbf{x}$, and the LMMSE of the true covariance minimises the posterior-mean error over linear maps, so the LMMSE floor bounds every linear estimator; and the Gaussian quasi-likelihood of noised data depends on the data only through their sample covariance, whose expectation is $e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}$, so a linear-Gaussian kernel fitted by maximum likelihood is consistent for (ρ, q) and lands on that floor. Proposition 1 thus turns the gap between S and S^{G} into an interpretable object: it is precisely what any estimator that carries only second-order structure—a linear-Gaussian kernel, a linear denoiser, or the Kalman smoother—must leave on the table.

BP with Gaussian messages is the Kalman smoother. On a Gaussian chain everything is available in closed form, which makes it the one place where grid BP can be checked against two independent exact computations.

Proposition 2 (The Gaussian chain: score, smoother and sum-product coincide). Let the driving noise of Equation (3) be Gaussian, so that $\mathbf{a} \sim \mathcal{N}(0, \boldsymbol{\Sigma}_0)$ with $(\boldsymbol{\Sigma}_0)_{ij} = \rho^{|i-j|}$, and let $\mathbf{x}$ follow Equation (2). Then (i) $S(\mathbf{x},t) = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$ and $\langle\mathbf{a}\rangle_{x,t} = e^{-t}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, with $\boldsymbol{\Sigma}_t = e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}$, and Tweedie's identity holds between them; (ii) every message of Equation (5) is Gaussian, $\overrightarrow{m}_i$ is the Kalman filter's prediction density $p(a_i \mid x_{1:i-1})$ and $\overrightarrow{m}_i\ell_t$ its filtering density, and the site marginals $\propto \overrightarrow{m}_i\ell_t\overleftarrow{m}_i$ are those of the Rauch–Tung–Striebel smoother, whose means are therefore $\langle\mathbf{a}\rangle_{x,t}$ of (i).

The proof is in Appendix A. The forward sweep is the Kalman filter and the full pass is its smoother, so a Gaussian BP is a linear map of the data and has little use beyond serving as a reference; this is what makes it a sanity check for any approximation of BP, grid BP included. Figure 4 measures it. A Rauch–Tung–Striebel smoother written without the BP code agrees with the closed form to 1.7×10^{-15}, and grid BP at the operating grid agrees with both to 1.9×10^{-13} at every one of the 12 noise levels. Refining the grid converges faster than any power of the spacing, as the trapezoid rule does on Gaussian integrands, and reaches 10^{-10} by $N_{\mathrm{g}} = 81$ at the smallest level, where the likelihood is narrowest, and by $N_{\mathrm{g}} = 41$ at the largest; widening the domain at the operating spacing removes the truncation error, which is 9.9×10^{-2} at $A = 3$.

What does not carry over. Moment-projected BP run on the grid is off by up to 2.6×10^{-1} even on the Gaussian chain: its backward messages are Gaussians whose variance grows like e^{2t}, and the grid truncates them at $\pm A$; the proposition concerns functional messages, and the LMMSE floors reported here are computed in closed form. On the ϕ^4 field the projected recursion is not well posed at all, because the backward messages are nearly flat (a fraction 0.42 of their mass lies in the outer 5% of the grid on the base cell and 0.98 or more on the others), so the Gaussian arm of every field comparison is the covariance twin itself, computed in closed form (Section 7). We also do not call the Gaussian pass a Kalman filter: it smooths.

5. Learning the local law through noise

Now drop the assumption that K is known. We observe M noised sequences $\mathbf{x}^\mu$, each at one level t_μ, never the clean chain, and maximise the marginal log-likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_t[t_\mu, \theta](\mathbf{x}^\mu)$ of a parameterised kernel K_θ. Because Section 3 computes this likelihood and its gradient exactly, learning the kernel and inferring the posterior are the same recursion.

The exact gradient from one sweep. Because only the transition factors depend on θ, Fisher's identity (Cappé et al., 2005) gives, on the grid,

$$\nabla_\theta\mathcal{L}(\theta) = \big\langle \boldsymbol{\Xi}(\theta),\, \nabla_\theta \log K_\theta \big\rangle,$$
$$\boldsymbol{\Xi}_{kl} = \sum_\mu \sum_{i\geq 2} P(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu), \qquad (6)$$

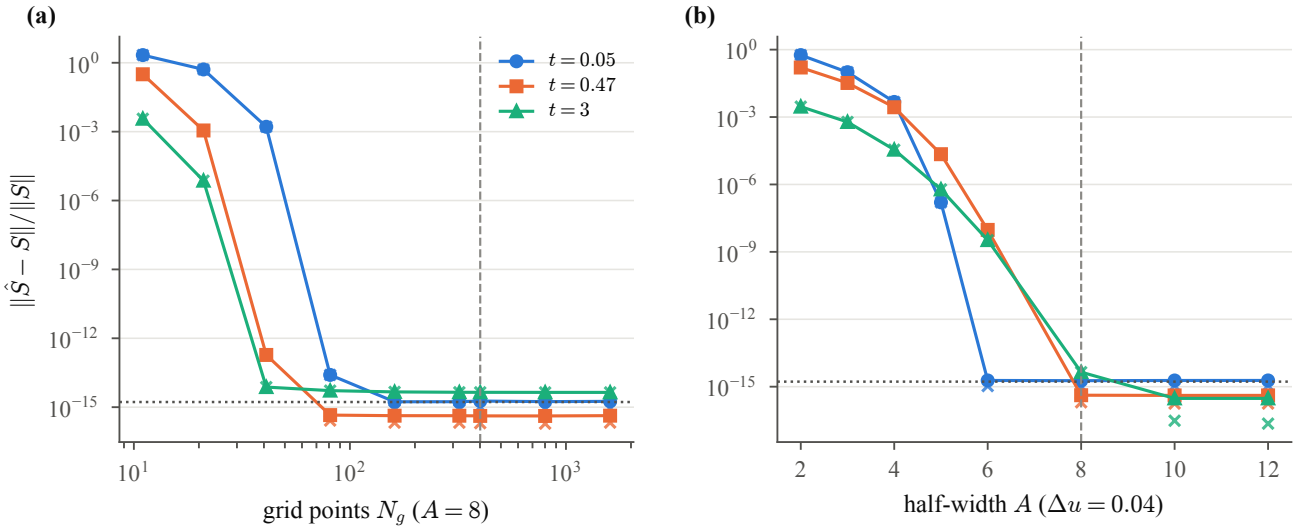


Figure 4. The Gaussian sanity check. Pooled relative score error of grid BP against the closed-form score (lines) and against an independent Kalman–Rauch–Tung–Striebel smoother (crosses) on the Gaussian chain, at the smallest, a middle and the largest noise level. (a) Refining the grid at $A = 8$; (b) widening the domain at the operating spacing $\Delta u = 0.04$. The dotted line is the smoother's own agreement with the closed form; the dashed vertical lines mark the operating point $(N_g, A) = (401, 8)$.

where the expected transition mass Ξ is read off the pairwise marginals that the same forward–backward pass returns. No differentiation through the recursion is needed, and the identity is verified against finite differences and against brute-force enumeration of the evidence on short chains (tests/test_em_bp.py, tests/test_fisher_bruteforce.py). Ξ is also all the M-step of EM needs, so EM alternates the two halves of Figure 2: run BP with the current kernel to get Ξ (E-step), refit the kernel to Ξ (M-step). The estimator sees one frozen noised view per sequence and never a clean trajectory; each reported fit selects its iteration count on a validation bundle disjoint from the test bundle, by the held-out risk against the exact score or, with no true score anywhere, by the held-out one-view log-evidence (Appendix C).

The kernel families. The M-step depends on how the kernel is parameterised, and the choice is the main modelling decision.

- Mixture kernel (25 numbers). For the linear chain, $K_\theta(a' \mid a) = \sum_{c=1}^{C} \pi_c \mathcal{N}\big(a' - \rho a - \nu_c, s_c^2\big)$: a linear drift with driving noise an C-component Gaussian mixture. $C = 1$ is the linear-Gaussian kernel and recovers the Gaussian score (Proposition 1); $C = 8$ is the paper's estimator. For ρ it is the belief-weighted Yule–Walker pair $\rho = S_{01}/S_{00}$, $q = (S_{11} - \rho S_{01})/W$ with $S_{00} = \sum_{kl} \Xi_{kl} u_l^2$, $S_{01} = \sum_{kl} \Xi_{kl} u_k u_l$, $S_{11} = \sum_{kl} \Xi_{kl} u_k^2$ and $W = \sum_{kl} \Xi_{kl}$; the complete-data problem for the mixture carries a second latent label and we take conditional-maximisation sweeps, a generalised EM whose ascent is monotone (Dempster et al., 1977; Meng & Rubin, 1993).
- Tilted-walk kernels (7, 25 numbers). For a nonlinear drift, $K_\theta(a' \mid a) = \mathcal{N}(a'; a, \gamma)\, e^{B(a')\cdot\theta}/Z_\theta(a)$, a Gaussian step with a learned tilt of the arrival point, B a Legendre or cubic-spline basis. It contains the field's kernel Equation (4) exactly, and the M-step objective is concave in θ at fixed γ.
- The site energy (5 numbers). Every tilted walk is reversible with the symmetric pair energy $W(a, a') = (a'-a)^2/2\gamma + f(a) + f(a')$, so the directed kernel and the pair energy are one model class (Appendix L). Parameterising f as a quartic polynomial and γ gives five numbers, exactly specified for the ϕ^4 field, and a fitted kernel reads as physics: the temperature $\hat{\Theta} = \hat{\gamma} s^2/h$, the barrier height and the well position.
- Unrestricted and neural kernels (100 and 1 516 numbers). A log-linear kernel with a free two-dimensional spline for $\log K$, which assumes no Gaussian bond, and a mixture density network (MDN) that maps a to a Gaussian mixture for a'. Both are fitted by the same EM; they are the tools for asking whether a simple family is wide enough (Section 9).

Resolution and capacity. A flexible kernel can learn narrow features, and the grid must resolve them: with large C and many sequences the mixture's narrow-

est component can be finer than a coarse grid's spacing, which we measure through a "resolved" flag (Section 8). The structure is not free either: $C = 1$ cannot see fourth-order structure and a large kernel overfits, which is why the tilted and site-energy families have few parameters.

6. A controlled laboratory: the linear chain with non-Gaussian noise

The linear chain of Equation (3) is the cleanest place to ask what structure is worth, because its covariance is fixed by two numbers and everything beyond it by the noise law, and because its exact score is available. We ask three things: what diffusion noise erases, and when (Section 6.1); whether the kernel can be learned through that noise (Section 6.2); and what explicit structure buys against networks (Section 6.3). Where it stops is Section 9. All comparisons use the frozen configuration of Appendix C: Laplace noise, $\rho = 0.85$, $L = 32$, twelve log-spaced levels on $[0.05, 3]$, and the schedule-pooled relative score error against the tree-exact reference. Paired seeds are the inferential unit, sixteen at the frozen configuration and eight in the sweeps across regimes; every selection is made on development seeds or on a validation bundle that never enters a reported number. Three information interfaces are distinguished throughout: (A) the standard denoising recipe, clean sequences with a fresh corruption at every step; (B) clean sequences with one fixed corruption each; (C) one noised view per sequence and nothing else, which is what EM–BP sees. Three reference points bracket every estimator (Appendix D): the isotropic score $-\mathbf{x}$ at 0.670, the LMMSE oracle with the true covariance at 0.168, the floor of every second-order estimator by Proposition 1, and the exact score at zero.

6.1. What noise erases: attenuation by statistical order

This subsection is the supporting theory: it says where non-Gaussian structure is lost, which is where the structured estimator can gain, and it does not enter the exactness of the method of Section 3.

Write $\mathcal{N}$ for the standard Gaussian on $\mathbb{R}^L$, He_k for the probabilists' Hermite polynomials and $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$ for their products, of degree $|\nu|$, with $\mathbb{E}_{\mathcal{N}}[H_\nu H_\mu] = \nu!\,\delta_{\nu\mu}$. The channel Equation (2) acts on Hermite moments diagonally.

Theorem 3 (Hermite attenuation). For every multi-index ν with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$, $\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)] = e^{-|\nu|t}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]$.

Proof. Mehler's identity $\mathbb{E}_z[\mathrm{He}_n(\alpha a + \sqrt{1-\alpha^2}\,z)] = \alpha^n \mathrm{He}_n(a)$ for $z \sim \mathcal{N}(0,1)$ follows from the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^n/n!$; coordinates are independent given $\mathbf{a}$, and $\alpha = e^{-t}$. □

Nothing here requires $P_0/\mathcal{N} \in L^2(\mathcal{N})$—which fails for the Laplace chain and, because $\lambda_{\max}(\mathbf{\Sigma}_0) > 2$, for the Gaussian chain at $\rho = 0.85$—only finite moments. Two consequences are exact at every t: cumulants of order $m \geq 3$ scale as e^{-mt} (Gaussian noise has none), and the degree-4 Hermite moments of two variance-matched laws differ exactly by their fourth-cumulant tensors κ_4, attenuated by e^{-4t}. Three clocks. For a feature first visible at Hermite degree m, its amplitude decays as e^{-mt} (exact, Theorem 3); the Fisher information a sequence carries about it, a squared amplitude, as e^{-2mt} (Theorems 5 and 6); and the number of sequences needed for a fixed local precision grows as e^{2mt} (Corollary 7). Correlation lives at degree two: e^{-2t}, e^{-4t}, e^{4t}. Non-Gaussian shape first appears at degree four: e^{-4t}, e^{-8t}, e^{8t}. Diffusion noise is a low-pass filter on statistical order.

The consequence for scores and for learning is an expansion in the signal-to-noise ratio e^{-2t}. Let $s = e^{-t}\mathbf{x}/\Delta_t$, $b = e^{-2t}/\Delta_t$ and $\Lambda(s,b) = \log \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$, so that $\log P_t(\mathbf{x}) = \log \mathcal{N}_{\Delta_t}(\mathbf{x}) + \Lambda(s,b)$ and $S = -\mathbf{x}/\Delta_t + (e^{-t}/\Delta_t)\nabla_s \Lambda$ exactly. Grading Λ by order in e^{-t}, its second-order part is the same for P_0 and its covariance twin, and its fourth-order part differs by a term linear in κ_4. The expansion is legitimate when Λ is finite in a neighbourhood of the origin, which is the one hypothesis the theorem needs:

Assumption 4 (Exponential moment). $\mathbb{E}_{P_0} e^{\lambda|\mathbf{a}|} < \infty$ for some $\lambda > 0$.

Every driving-noise law used below satisfies it except Student-t, for which $\Lambda(s,0) = +\infty$ at every $s \neq 0$; that family is discussed after Theorem 6.

Theorem 5 (Order-by-order information decay). Let P_0 satisfy Assumption 4 with covariance $\mathbf{\Sigma}_0$, let S^{G} be the score of its covariance twin, and fix L. As $t \to \infty$,

$$\mathbb{E}_{P_t}\|S - S^{\mathrm{G}}\|^2 = \tfrac{1}{6}\|\kappa_4\|_F^2\, e^{-8t}(1+\epsilon_t), \tag{7}$$

$$I_t(\theta) = \tfrac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2\, e^{-4t}(1+\epsilon_t) \text{ if } \partial_\theta \mathbf{\Sigma}_0 \neq 0, \tag{8}$$

$$I_t(\theta) = \tfrac{1}{24}\|\partial_\theta \kappa_4\|_F^2\, e^{-8t}(1+\epsilon_t) \text{ if } \partial_\theta \mathbf{\Sigma}_0 = 0, \tag{9}$$

the last with $\partial_\theta \kappa_4 \neq 0$, and with $\epsilon_t = O(e^{-2t})$, where $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ is the Fisher information one noised sequence carries about a parameter θ of P_0, and $\|\cdot\|_F$ is the Frobenius norm of the matrix or symmetric 4-tensor over ordered indices. For a Gaussian

chain Equation (8) is exact at every t with $I_t = \frac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_t\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_t)$ and $\epsilon_t = O(e^{-2t}\lambda_{\max}(\boldsymbol{\Sigma}_0))$ explicitly.

The information statements Equations (8) and (9) are the diffusion-time form of a principle established for low-SNR latent-variable models: the log-likelihood expands in powers of the SNR with the order-k term carrying k-th-moment information (Katsevich & Bandeira, 2023; Perry et al., 2019; Bandeira et al., 2023), and Fisher information about directions first identifiable at moment order k scales as SNR^k (Balanov et al., 2026); with $\mathrm{SNR}_t \simeq e^{-2t}$ these are Equations (8) and (9). The cumulant attenuation behind Theorem 3 is also written down for the forward process by Aarts et al. (2025). What this paper adds is the score-level statement Equation (7) with its constant, made interpretable by Proposition 1; the general order and its constant below; the hypothesis under which the expansion is a theorem and what happens outside it; and the exact verification and learning consequences that follow. The proof (Appendix A) expands Λ to fourth order on the bulk $\{|s| \le \lambda/2\}$, where Assumption 4 makes it analytic with uniformly bounded tilted moments, takes the $\mathcal{N}$-second moment of the leading Hermite term by Wick's theorem, and bounds the tail, whose probability is $\exp(-c\,e^{2t})$ under Assumption 4; the scalar leading term of Equation (7) is $(\kappa_4/6)\mathrm{He}_3(x)e^{-4t}$. The leading orders of a degree-2 and a degree-4 coordinate do not couple, because He_2 and He_4 are $\mathcal{N}$-orthogonal, so the efficient information of a shape coordinate after lower-degree nuisance coordinates are estimated has the same rate and constant as its raw information; a nuisance of the same degree costs an $O(1)$ factor.

Theorem 6 (General order). Let P_0^θ satisfy Assumption 4 uniformly near θ_0, and let m be the smallest degree at which $\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})] \neq 0$ for some $|\nu| = m$. Then, at fixed L,

$$I_t(\theta) = e^{-2mt} \sum_{|\nu|=m} \frac{\big(\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})]\big)^2}{\nu!}\big(1 + O(e^{-2t})\big).$$

The constant is the Hermite norm of the derivative of the degree-m moment tensor; for $m = 2$ it is $\frac{1}{2}\|\partial_\theta\boldsymbol{\Sigma}_0\|_F^2$ and for $m = 4$ at fixed covariance $\frac{1}{24}\|\partial_\theta\kappa_4\|_F^2$, since the multiset ν collects $4!/\nu!$ ordered index tuples. A direction with no non-vanishing Hermite footprint is not identifiable at any noise level; under Assumption 4 the law is determined by its moments, so such a direction is null.

Corollary 7 (Sample complexity by order). Fix t and let θ be a coordinate of lowest degree m with efficient information $I_t^{\mathrm{eff}}(\theta)$. For M i.i.d. sequences observed at level t and any estimator sequence, the local asymptotic minimax risk satisfies $\lim_{c\to\infty}\liminf_M \sup_{|\theta'-\theta|\le c/\sqrt{M}} M\,\mathbb{E}_{\theta'}(\hat\theta - \theta')^2 \ge 1/I_t^{\mathrm{eff}}(\theta) = \Theta(e^{2mt})$, and maximum likelihood attains it. A covariance coordinate therefore needs $M \propto e^{4t}$ sequences for a fixed local precision, a shape coordinate $M \propto e^{8t}$; the ratio of the two informations is $\propto e^{-4t}$.

This is Hájek–Le Cam for a regular parametric model at fixed t; it says nothing uniform in t (the limits do not commute), nothing about absolute precision for a bounded coordinate once $1/\sqrt{MI_t}$ exceeds its range, and nothing about an optimiser at a finite budget. All three caveats are visible in Section 6.2.

Where the asymptote begins. For the Gaussian chain the expansion parameter is $e^{-2t}\lambda_{\max}(\boldsymbol{\Sigma}_0)$, with $\lambda_{\max} \le (1+\rho)/(1-\rho)$, and $t_* = \frac{1}{2}\log\lambda_{\max}(\boldsymbol{\Sigma}_0)$ is its scale, exact for the Gaussian chain in the sense that $P_t/\mathcal{N} \in L^2(\mathcal{N})$ iff $e^{2t} > \lambda_{\max} - 1$, that is for $t > \frac{1}{2}\log(\lambda_{\max} - 1)$, just below t_*; below t_* the long-wavelength covariance modes are still of order one and the rates are shallower. For non-Gaussian laws the same scale is a heuristic that the measurements below support. At $\rho = 0.85$, $t_* = 1.17$: a strongly correlated chain keeps its correlation information longer, and any measurement stopped near t_* sees slopes short of -4 and -8.

Heavy tails. Student-t_ν driving noise violates Assumption 4. For a regularly varying tail of index ν the expected squared gap acquires a term of order $e^{-\nu t}$ carried by sequences with $|\mathbf{a}| \gtrsim e^t$, whose probability is itself of order $e^{-\nu t}$: in expectation the tail term leads once it exceeds $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, and on any sample of practical size the Hermite law is all that is seen, because the events that carry the tail term are never drawn. Both halves are visible in the numbers (Appendix K): for the scalar t_5 law an untruncated quadrature gives a local slope of -5.28 on the last interval and a ratio to the Hermite term of 9416 at $t = 5$, while the t_5 chain sampled with 8192 sequences on a domain wide enough to hold them sits at 1.03 $[0.99, 1.09]$ of the constant at $t = 4$ (Figure 5c). We state the heavy-tailed rate as an observation supported by quadrature, not as a theorem; the moment condition of Theorem 5 is not merely technical.

Verification with the exact score. Figure 5 tests Theorem 5 where nothing has to be approximated. For the Gaussian chain the closed form gives slope -3.97 above t_* at $\rho = 0.85$; for the non-Gaussian chain the in-

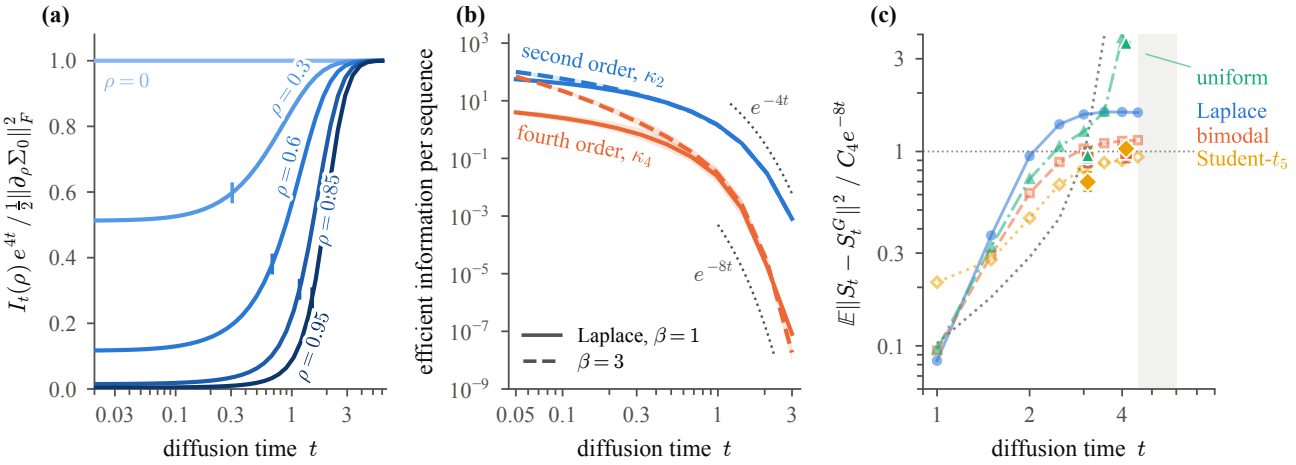


Figure 5. What survives diffusion noise. (a) Exact Fisher information of the Gaussian chain about ρ, scaled by its predicted asymptote $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 e^{-4t}$, for five ρ; every curve reaches 1, and the tick on each marks $t_*(\rho)$. (b) Efficient information per noised sequence about the second-order cumulant $\kappa_2 = q$ and the fourth-order cumulant $\kappa_4 = \gamma_2(\beta) q^2$ of the driving noise of the generalised-Gaussian chain (γ_2 its excess kurtosis), for the Laplace member ($\beta = 1$, solid) and for $\beta = 3$ (dashed), from tree-exact inference, Fisher's identity and the change of coordinates from (ρ, β) to the cumulants (Appendix K); 512 sequences $\times$ 8 replicates (bands: replicate range); guides e^{-4t} and e^{-8t}. Above t_* the fitted slopes at $\beta = 3$ are -3.99 (κ_2) and -8.01 (κ_4); the Laplace member's fourth-order curve reaches the estimator's numerical floor from $t \approx 3$ on. The coordinates are cumulants, not the shape parameter β. (c) The price of second-order closure: $\mathbb{E}\|S - S^{\mathrm{G}}\|^2$ measured by tree-exact BP, divided by the predicted $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, for four driving-noise families on a domain wide enough for their samples (Appendix B). Thin curves: 512 sequences, paired across levels; bold markers with bars: the constant from 8192 sequences with a bootstrap interval at $t = 3$ and 4; grey dotted: the same Student-t_5 sequences on the operating $(801, 8)$ domain, where the truncated prior drives the reference. The shaded band is where the double-precision floor of the posterior means enters; the uniform family, with the smallest constant, reaches it first. Laplace, the bimodal mixture and Student-t_5 reach the constant.

formation about the fourth cumulant, estimated from per-sequence scores through the tree-exact recursion, falls with slope -7.94 ($\beta = 2$) and -8.01 ($\beta = 3$) above t_* against the predicted -8, and the information about the second cumulant with slope -3.99 against -4. These are the order-by-order coordinates of the theorem; the same data plotted against the shape parameter β would misstate the fourth-order information by a factor between 1.6 and 1526, because β barely moves any cumulant near the Gaussian. The constants are verified at two levels of precision: in the scalar case an untruncated quadrature reproduces the shape constant of Equation (9) to within 0.6% and the covariance constant of Equation (8) to five digits (Appendix K), while on the chain the outer-product estimate of the shape constant reaches it only to within twenty per cent before its quadrature floor; the closure gap of the Laplace chain decays with slope -7.93 and reproduces $\frac{1}{6}\|\kappa_4\|_F^2$ to 0.99 $[0.91, 1.08]$ at $t = 4$ from 8192 sequences (the per-sequence gap is heavy-tailed, which is why so many are needed). The theorem also explains a measurement the previous literature on this model reported as a discrepancy: slopes of about -3 and -6 obtained over $t \le 1.6$ are the pre-asymptotic values below $t_* = 1.17$, not a failure of the law. Finally, Equation (7) says what the covariance twin lacks: adding the degree-4 Hermite term back, the first Edgeworth correction with the true $\mathbf{\Sigma}_0$ and κ_4, closes 0.961 of the squared gap at $t = 0.68$ and all of it from $t = 2$ on every family with an exponential moment, and diverges below $t = 0.15$ (Appendix K); above the crossover a score built from the first four moments is the score.

What this says about score learning. An estimator that carries only second-order structure cannot go below the LMMSE floor $\|S - S^{\mathrm{G}}\|$, which on the Laplace chain at $\rho = 0.85$ is 0.168 of the score energy pooled over the twelve-level schedule and 0.28 at the lowest level (Appendix D). A learner that recovers the fourth-order structure has, by Corollary 7, e^{4t} fewer effective samples per sequence at level t than one that recovers only covariance: the twelve-level schedule is, for shape, effectively a four-level one. Both facts organise the experiments of Section 6.

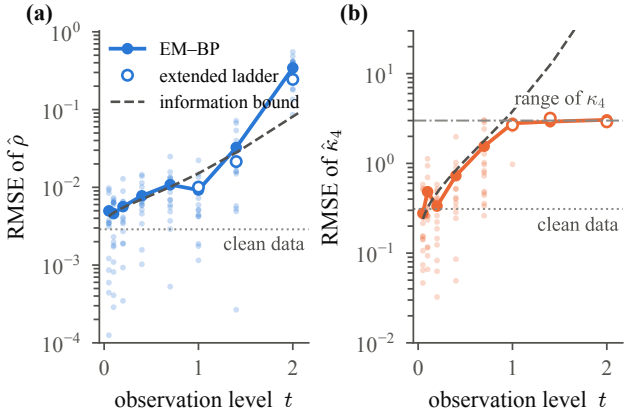


Figure 6. Learning through noise. Recovery of the correlation coordinate and of the driving-noise shape from $M = 512$ sequences observed at a single level t, sixteen replicates (points: individual fits; line: RMSE), against the Cramér–Rao reference (dashed), the clean-data floor (dotted) and, for the shape, the coordinate's range. The fits are unimodal and at the reference at the lowest levels, split into a shrunk cluster and Gaussian-collapsed fits from $t = 0.7$, and collapse where the reference crosses the range. Hollow markers: the same fits with the em ladder extended to 8000 iterations (e06c).

6.2. Learning the shape and the correlation through noise

The information curve predicts what is learned at each level. Corollary 7 makes a prediction about learning that the exact score does not need: fitting from M sequences all observed at level t, the shape of the driving-noise law stops being identifiable at the level where its Cramér–Rao reference crosses the coordinate's range, while the correlation survives to the crossover. This is also the arithmetic behind the one-view protocol: of twelve levels spread over $t \in [0.05, 3]$, only the lowest four carry appreciable information about shape.

Figure 6 measures both with $M = 512$ Laplace-chain sequences at $\rho = 0.85$ and 16 replicates per level, against the information reference of Section 6.1: the Cramér–Rao bound $1/\sqrt{M\, I_t}$ for the correlation and, by the chain rule at the Laplace member of the shape family, for the excess kurtosis. The reference is for an unbiased estimator in a one-parameter family and the estimator is an eight-component mixture with its iteration count selected on validation, so the comparison is a yardstick, read level by level. Up to $t = 0.4$ every fit recovers the shape and the dispersion of $\widehat{\kappa_4}$ across replicates is within the sampling error of the reference (0.69 against 0.838 at $t = 0.4$; every level in Appendix K). At $t = 0.7$ the fits split, 4 of 16 collapsing to Gaussian driving noise and the rest shrunk; at $t = 1$, 15 of 16 collapse, and the reference has crossed the full range of the coordinate at $t = 0.98$: the level at which the information runs out is the level at which the estimator stops estimating. The correlation is recovered to 0.0092 at $t = 1$ against a reference of 0.0151 and is lost only past the crossover $t_* = 1.17$, 0.0325 at $t = 1.4$ and 0.3430 at $t = 2$, of which part is budget rather than information: with the em ladder extended to 8000 iterations on the same seeds and data (e06c; hollow markers) the two fall to 0.0215 and 0.2441 while nothing below $t = 1$ moves by more than 0.0009; past the crossover em's contraction slows with the information it has. The same coordinates estimated by the method of moments—the lag-one product for ρ and the per-site fourth cumulant divided by e^{-4t} for κ_4, whose marginal statistics are unbiased by Theorem 3 up to the $O(e^{4t}/M)$ plug-in bias of a sample cumulant—are 5–6 times worse for ρ and 2.4–5.5 times worse for κ_4 at the informative levels, never collapse, and show the asymptotic growth where the estimate is already useless (log-slopes 1.83 and 3.73 on $t \geq 1$ against 2 and 4; Appendix K): exact inference buys the efficiency below the crossover, and at high noise the two estimators share the rate, since the likelihood score of a degree-m coordinate is itself a degree-m Hermite statistic, of which the lag-one and per-site statistics used here keep only a part. Below the crossover both errors grow more slowly than the asymptotic rates (fitted log-slopes 1.29 and 3.03 on $[0.2, 0.7]$ against 2 and 4), as Section 6.1 predicts; the asymptotic window lies beyond the reach of $M = 512$ sequences at this correlation, because the shape collapses before t_*. The clean-data fit gives the optimiser's own floor, 0.0029 and 0.310.

Table 1. Schedule-pooled relative score error on all sites, mean over 16 seeds (s.e. in the text macros; the isotropic reference is 0.669 at every size). Every arm is scored on identical test bundles. The last column is the same capacity fitted to the driving-noise law itself—the C-component mixture closest in Kullback–Leibler divergence, an infinite-data reference for the family rather than a bound; † marks fits whose narrowest component is under two grid cells.

	sequences M				
kernel	32	128	512	2048	$M = \infty$
LMMSE floor	0.167	0.167	0.167	0.167	0.167
$C = 1$	0.174	0.168	0.168	0.167	0.167
$C = 2$	0.084	0.050	0.035	0.028	0.039
$C = 4$	0.093	0.053	0.037	0.025	0.0057
$C = 8$	0.099	0.055	0.036	0.023	0.0022†
$C = 16$	0.100	0.056	0.036	0.022	0.0007†

6.3. What explicit structure buys

Capacity: the margin is fourth-order. Table 1 places every arm on identical test bundles. A linear-Gaussian kernel ($C = 1$) lands on the LMMSE floor, within sam-

pling error, at every size (0.174, 0.168, 0.168, 0.167 against the floor 0.167–0.167), as Proposition 1 requires; the mixture kernel goes far below it, reaching 0.023 at $M = 2048$, 7.9× under the floor. The paired $C = 8$ versus $C = 1$ difference is 0.144 (bootstrap $[0.141, 0.147]$) at $M = 2048$ and 0.075 ($[0.059, 0.090]$) already at $M = 32$, in favour of the mixture in 16 and 16 of 16 seeds. At $M = 2048$ the $C = 8$ arm still selects the top of its 1200-iteration ladder in 10 of 16 seeds, so its error there is an upper bound on what a longer ladder would give. The fitted driving noise excess kurtosis is 2.89 at $M = 2048$ against the true 3; 1 of 320 fitted cells fail the two-grid-cell resolution gate and are reported, not dropped. The last row of Table 1 is the family's infinite-data reference, the C-component mixture closest in Kullback–Leibler divergence to the Laplace driving noise: its $C = 1$ member scores 0.167, the LMMSE floor to three decimals, and at $C = 8$ it scores 0.0022, so the fitted kernel at $M = 2048$ sits 10.3× above what its own family can express, a data limit and not a capacity limit (Appendix F). An earlier study of this model concluded the opposite from a sweep that, it turned out, had fitted every arm to a single training chain (Appendix F); with independent chains and the checkpoint selected identically for every capacity the effect is unambiguous. Capacity buys exactly what the closure discards: across the generalised-Gaussian family at fixed covariance the paired gain of $C = 8$ over $C = 1$ is ordered by κ_4, 0.131 at the Laplace member and a small loss, −0.008, at the Gaussian member, where there is no fourth-order structure to buy (Appendix G, Figure 17). The contrast is not a property of the frozen point: in all 15 cells of a sweep over $\rho \in [0.3, 0.95]$ and $L \in [16, 64]$ the mixture beats the Gaussian kernel with the paired interval excluding zero, the paired difference $C = 8$ minus $C = 1$ running from −0.253 at the most favourable corner to −0.047 at the least, and in every cell the $C = 1$ arm lands on the LMMSE floor to 0.0099, which checks the whole apparatus away from the frozen configuration (Appendix H).

Baselines under one recipe. Table 2 compares six architectures trained under one recipe (interface A) with the configuration that won each family's screen on the development seeds (Appendix C). Weight sharing separates the fully connected network from the rest at every size (0.517 against 0.186 at $M = 32$); locality is the better prior when sequences are scarce, the window head being the best network at $M = 32$, and stops paying once they are not: the window head moves only from 0.121 to 0.118 between $M = 512$ and 2048 while the global architectures keep improving, the best reaching 0.060. That plateau is the architecture's and not the radius's: the best estimator of any kind that sees only a window of radius r is $\mathbb{E}[a_i \mid x_{i-r}, \ldots, x_{i+r}]$, computable by sum-product on the sub-chain, and at radius 4 its pooled error is 0.030 against the head's 0.118 (Appendix K, Figure 18). Given enough clean sequences the networks pass the second-order floor, 5 of six sitting below 0.167 at $M = 2048$ (Proposition 1); nothing intrinsic holds them at second order. The structured estimator is ahead at every size nonetheless, 0.023 against 0.060 at $M = 2048$, a paired factor 2.83 ($[2.37, 3.42]$, 16 of 16 seeds). That factor is an upper bound on the advantage at this interface, since the largest networks were still improving when their ladder ended (88% of the transformer cells at the last rung), and it is not information-matched: the networks see clean sequences and a fresh corruption at every step, EM–BP one noised view per sequence and no clean chain at all.

The comparison at matched information. Interface C is EM–BP's own: one noised view per sequence split across the twelve levels, no clean chain, and, since the true score is unavailable for selection there, the checkpoint chosen on the implicit score-matching objective of a held-out noisy bundle. Trained by implicit score matching on exactly that bundle, the window head reaches 0.165 at $M = 2048$ against EM–BP's 0.023, a paired factor 7.9 ($[6.6, 9.3]$, 16 of 16 seeds); the transformer reaches 0.421. Under interface B, the clean sequence supplied as a regression target but with one fixed corruption per sequence, the same head reaches 0.155, a factor 7.3: even given strictly more than EM–BP sees, one corruption per sequence is not enough. The factor that remains at equal information is what the factorisation and the exact E-step are worth, and it is larger than the factor at the standard interface, not smaller. EM–BP's own selection needs no true score either: choosing its iteration count by held-out one-view log-evidence gives 0.105 against 0.105 with oracle validation at $M = 32$, where the rung matters most, the two rules agreeing in 7 of 8 seeds (Appendix K).

The matched comparison away from the frozen point. The factor is not a property of the frozen corner. Re-running the matched comparison across $\rho \in \{0.5, 0.85, 0.95\}$ and $L \in \{32, 128\}$, with the networks re-screened inside each regime and 16 seeds per cell (Appendix J), EM–BP with its iteration count chosen by held-out evidence, so that no true score enters either arm, is ahead of the best implicit-score-matching network in all 6 cells and in every seed: by a paired factor 5.7 [4.9, 6.7] at $\rho = 0.5$, $L = 32$, and 8.3 [6.6, 10.3] at the frozen point, consistent with the sixteen-seed 7.9 above. The factor is smallest at $\rho = 0.5$ at

Table 2. All-site schedule-pooled relative score error, mean over the confirmatory seeds (± s.e. in the macros), every arm on identical test bundles. Oracles and EM–BP ($C = 8$) from the capacity experiment; networks under the common recipe with the screened configuration.

M	isotropic	LMMSE	MLP	window	TCN	bi-GRU	transf.	U-Net	EM–BP
32	0.669	0.167	0.517	0.186	0.223	0.201	0.231	0.227	0.099
128	0.669	0.167	0.391	0.141	0.139	0.121	0.151	0.150	0.055
512	0.669	0.167	0.276	0.121	0.096	0.078	0.097	0.095	0.036
2048	0.669	0.167	0.208	0.118	0.068	0.060	0.063	0.066	0.023

both lengths, where the stationary marginal carries the most fourth-order structure (1.80 against 0.15 at $\rho = 0.95$), which is the direction Section 6.1 gives; beyond $\rho = 0.85$ the two lengths part company, the factor rising further at $L = 32$ and falling back at $L = 128$, so we report the direction below $\rho = 0.85$ and no trend above it. Lengthening the chain does not collapse the advantage (4.6 to 8.0 at $L = 128$). Given the clean sequence as a regression target (interface B) the best network is still a factor 7.6 behind at the frozen point, and the factors are conservative throughout: the evidence-selected EM–BP selects the last rung of its ladder in 56 of 96 cells, so its error there is an upper bound. Against the second-order floor the sweep separates two questions that coincide at the frozen configuration. The best network's error changes little across it (0.089 to 0.196) while the floor falls steeply with ρ, so the network passes the floor where the floor is high ($(0.5, 32), (0.5, 128), (0.85, 128)$) and not where it is low ($(0.85, 32), (0.95, 32), (0.95, 128)$): at $\rho = 0.95$ it is above by $+0.110$ [$+0.108$, $+0.112$] at $L = 32$, against an exact radius-8 floor of 0.025 at that correlation. Where the covariance carries most of the score energy, one noised view per sequence is not enough for a network to recover even the covariance, while the estimator that knows the factorisation recovers it. The same cell with platykurtic driving noise gives the same answer: on the symmetric two-component mixture, whose excess kurtosis is negative at the same covariance (Appendix D), the paired factor is 8.8 [7.4, 10.2] (Appendix J). What the structure buys is fourth-order structure of either sign, not heavy tails.

Length extrapolation. Fitted at $L = 32$ and evaluated against the exact score of longer chains (Table 9), the EM–BP kernel holds 0.023 at $L = 32$ and 0.023 at 256: the law is local, so inference at any length is the same recursion. The convolutional and recurrent networks improve slightly, as longer chains have proportionally fewer boundary sites; the transformer with learned relative positions degrades from 0.063 to 0.703, a failure of the positional encoding beyond its training length rather than of the score it learned (Press et al., 2022; Kazemnejad et al., 2023).

Generation. Samples are drawn by the same reverse SDE (400 steps on a geometric grid, one Brownian path shared by every arm, 16 seeds) from the exact score, the fitted kernel, the LMMSE twin and the isotropic score, and compared with a reference sample of the true law at $t_{\min}$ (Table 8). The exact arm's log-likelihood gap, 0.070 nats per sequence at $M = 2048$, is the sampler's own discretisation error, the reference against which the other arms are read; the fitted kernel sits at 0.164, the LMMSE twin at 1.766 and the isotropic score at 46.295. The excess kurtosis of the generated sequences' one-step residuals at $t_{\min}$, where the residual noise lowers even the true law's value below the driving noise's 3, is 1.803 for the fitted kernel against 1.94 for the true law and -0.003 for the twin, which by construction generates Gaussian driving noise: what the second-order estimator discards in score error it discards in the samples. Because the kernel is known after learning, generation does not need the reverse process at all: one forward sweep from the learned kernel draws a sequence, and with the true kernel it samples at the Monte Carlo floor of exact draws while costing 731× less than a 200-step reverse SDE (Appendix I). A Gaussian score generates Gaussian sequences, whatever its accuracy at second order.

7. A physical benchmark: the one-dimensional ϕ^4 field

The linear chain tests the method on a model chosen for solvability. The ϕ^4 field of Section 2 tests it on a model chosen by physics: spatial snapshots of the stochastic Allen–Cahn equation, whose stationary law is a Gibbs measure with domains, kinks and a tunable correlation length. Because its lattice snapshots are exactly Markov (Equation (4)) and its transfer operator can be diagonalised, the exact score, the exact samples and the exact physical statistics are all available, so every estimator can be compared with the truth and with a physical reading. The comparison follows the four arms of the proposal that motivated it: a learned pair energy, Gaussian closure, a window CNN, and the exact oracle.

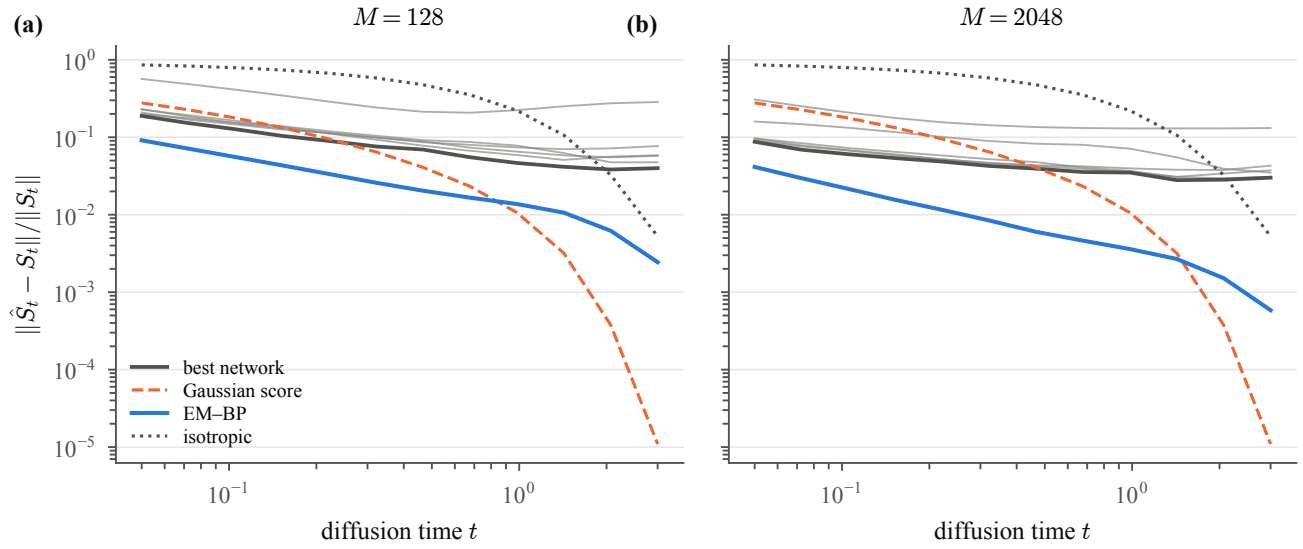


Figure 7. Where the gain is. Per-level relative score error against the exact score on the linear chain with Laplace noise, for $M = 128$ (a) and $M = 2048$ (b) training sequences: EM–BP (blue), the best of the six networks (thick grey; the other five thin), the Gaussian score (dashed) and the isotropic score (dotted). EM–BP is the best estimator at the lowest 8 (a) and 10 (b) of twelve levels. The Gaussian score, which is given the true covariance, overtakes it from $t = 0.98$ (a) and $t = 2.07$ (b) upward (the first level at which it is better, median over seeds), because nothing non-Gaussian is left to recover; no network is first at any level.

Table 3. The six cells of the field laboratory: temperature Θ, lattice spacing h (solved on the three hotter cells to hold the correlation length fixed), chain length L, correlation length ξ in sites, excess kurtosis of the marginal (kurt.), and the schedule-pooled relative score error of the best linear estimator (floor) and of the isotropic score (iso.), mean of two replicates.

Θ	h	L	ξ	kurt.	floor	iso.
0.3	1.000	32	6.1	−1.35	0.269	0.674
0.5	0.419	32	6.2	−0.88	0.115	0.657
1.0	0.226	32	6.2	−0.59	0.058	0.657
2.0	0.150	32	6.2	−0.46	0.041	0.656
0.3	0.500	64	11.5	−1.25	0.162	0.714
0.3	0.250	128	22.5	−1.22	0.098	0.750

Cells. Two knobs set the chain, the temperature Θ and the lattice spacing h, and we run six cells (Table 3). The base cell ($\Theta = 0.3$, $h = 1$, $L = 32$) is cold enough for domains. Three hotter cells hold the correlation length in sites fixed by solving for h at each Θ, so they compare the field with the linear-chain laboratory at matched second-order structure. Two refinements of the base cell hold the physical length fixed ($h = 0.5$ with 64 sites, $h = 0.25$ with 128), so the correlation length in sites grows like $1/h$ (from 6.1 to 22.5). The linear floor of every cell is the risk of the best linear (Gaussian) estimator (Proposition 1); it is large where the field is cold and nonlinear.

Temperature and domains. At fixed spacing the physics is monotone in the temperature (Figure 10): lowering Θ lengthens domains (the correlation length in sites grows from 1.06 at $\Theta = 2$ to 19.5 at $\Theta = 0.2$, exponentially in $1/\Theta$ with a slope that approaches the kink energy $E_{\text{kink}} = 2\sqrt{2}/3$ as the field cools; Appendix L), sharpens the bimodality (coefficient from 0.45 to 0.73), and makes the linear floor higher (0.184 at $\Theta = 2$ to 0.296 at $\Theta = 0.2$). The raw count of sign changes per unit length is not a physical kink density (a rough field crosses zero inside a domain, and more finely on finer lattices), so we count kinks after merging runs shorter than one kink width, which is lattice-independent (Appendix L). There is no transition: the crossover is smooth.

Arms and protocol. Every estimator learns from noised sequences only. EM–BP with five kernel families (Section 5): the paper's mixture (25 numbers), a spline tilt (25), a Legendre tilt (7), the quartic site energy (5), and an MDN (1 516); a window CNN and a transformer, each trained under two recipes, implicit score matching on one noised view per sequence (ISM, EM–BP's own interface) and denoising score matching on clean targets with one fixed corruption (DSM, which gives the network strictly more); the linear floor; and the exact oracle. Sizes are $M \in \{32, 128, 512, 2048\}$ for the kernels and $\{128, 2048\}$ for the networks, sixteen seeds, the same twelve levels, and the held-out risk against the exact score selects EM's

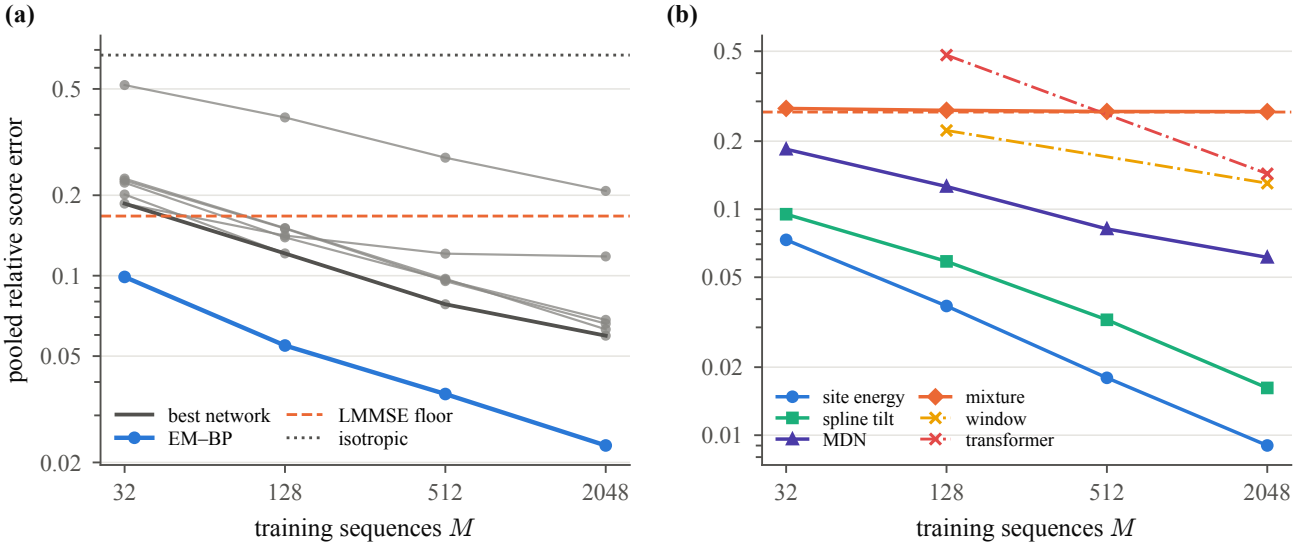


Figure 8. Sample efficiency. Pooled relative score error against the number of training sequences. (a) The linear chain with Laplace noise: EM–BP ($C = 8$), the six networks (best in dark grey) and the LMMSE floor (dashed). (b) The ϕ^4 field at $\Theta = 0.3$, $h = 1$: the site energy, a spline-tilted kernel, an MDN kernel, the paper's mixture kernel and two networks, against the field's linear floor (dashed).

stopping point for every arm alike.

The site energy wins on every cell; the paper's kernel does not leave the linear floor. Table 4 is the central result. At $M = 2048$ the five-parameter site energy reaches 0.032–0.169 of the linear floor across the six cells, the best method on every cell at every size on average, and 0.41–0.57 of the 25-parameter spline's risk; the spline is next, and the networks, whichever recipe, are above the linear floor on every cell with $\Theta \geq 0.5$ and at least 4.1 times the spline's risk on every cell at both network sizes. The paper's own mixture kernel is the instructive failure: it sits on the linear floor (1.004–1.021 times it at $M = 2048$) on every cell, because its expected next value is a linear function of the current one, whereas the field's is not (Appendix L); with more data its gap to the spline widens rather than closes (its risk is 9.5 times the spline's at $M = 512$ and 18.2 times at 2048 on the base cell). A kernel whose family contains the physics pays nothing for it, a kernel whose family misses it never recovers; the choice of kernel family is the modelling decision, and for a gradient flow the right family has five numbers.

The fitted kernel reads as physics. The site energy is the physicist's pair energy, so its parameters are physical. Fitted through noise to $M = 2048$ sequences, the temperature is recovered to within 10% in 96 of 96 fits and the barrier height to within 20% in 78 of 80 fits on the five cells where the registered reading asks for it (Table 11); at $\Theta = 2$, where the barrier is small against the thermal energy, its error falls like $M^{-0.54}$ and every one of sixteen seeds is within 20% at $M = 32\,768$ (Appendix L), so it is a sample-size limit and not a floor.

Generated fields are physically right. A score can be accurate and still generate wrong fields, so we sample from each fitted kernel by one forward sweep and compare physical statistics with exact draws (Table 5). Samples from the site energy and the spline reproduce the coarse kink density within 2.1% and 1.1% of the truth on every cell, the mean domain length likewise, and the whole distribution of domain lengths to 5.6% and 10.4% at every quantile, statistically indistinguishable from exact samples. The mixture's samples have 1.41–1.42 times too many kinks on the cold cells and domains 14%–18% too short at the median: a linear drift cannot hold a domain, which is the smearing of the Gaussian closure seen in the samples.

Refinement: what a fixed receptive field costs. Because the correlation length in sites grows like $1/h$, a denoiser with a fixed window should lose ground as the lattice is refined and BP should not, which is the proposal's sharp prediction. The window CNN's radius is screened per cell (2, 4 and 8 sites at $h = 1, \frac{1}{2}, \frac{1}{4}$, a constant physical reach, the last at the largest radius offered). Its absolute risk falls (0.149, 0.118 at $h = 1$, 0.25), but the structured kernel's falls faster: the window's risk relative to the spline grows from 9.2 to 17.6 and relative to the linear floor from 0.55

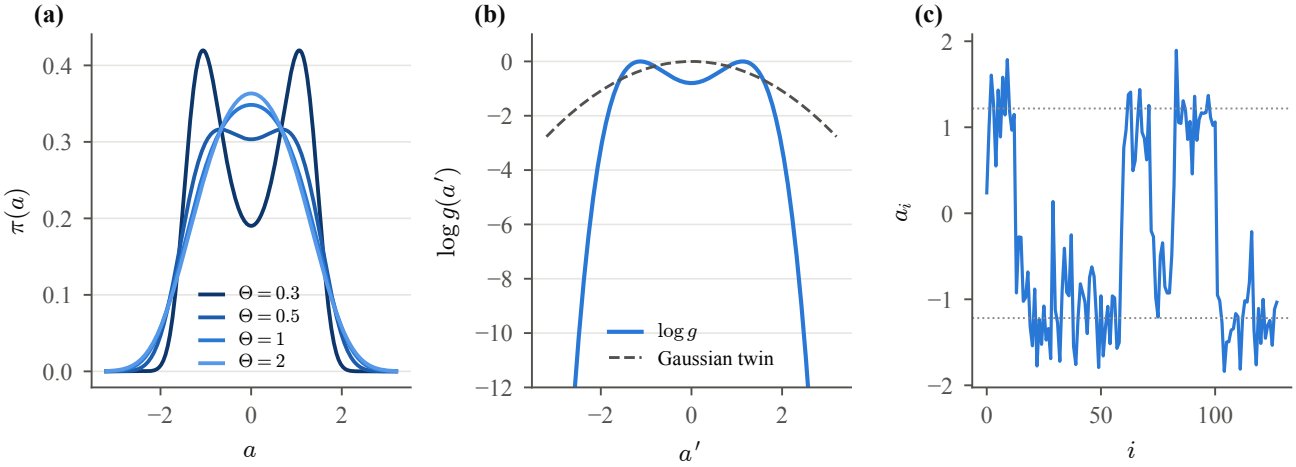


Figure 9. The field as a Markov chain. (a) Stationary density of the standardised field at the four temperatures of the sweep of Table 3: two humps when cold, a single hump when hot. (b) The log of the tilt g of Equation (4) against the log of the Gaussian tilt that would give the covariance twin; the field's tilt is bimodal, which a linear-Gaussian kernel cannot express. (c) An exact sample of 128 sites at the base cell's temperature and spacing; dotted lines mark the two wells.

to 1.20, so on the finest lattice the window is worse than the Gaussian closure. Figure 11 puts this beside cost: the networks own the cheap end of the runtime–accuracy frontier (0.004–0.32 ms per sequence against 0.51–2.0 ms for grid BP, which is shared by every BP-based arm) and the structured kernels the accurate end, with nothing between them.

Speciation: when does the denoiser commit to a domain? The proposal that motivated this benchmark asks to watch the denoiser commit to whole domains as the noise is removed. For a noised sequence the exact posterior gives each site $p_i = P(a_i > 0 \mid \mathbf{x})$, and its commitment is $|2p_i - 1|$, zero when the two wells are equally likely and one when the site has chosen. Averaged over sites, the commitment falls from one at $t = 0$ to zero at high noise; we call t_c the level where it crosses one half and compare it with the Gaussian twin, whose commitment is available in closed form (Appendix L). The posterior is calibrated, so the commitment is the probability of guessing the sign right (the two agree to 0.003 on average over levels). Figure 12 gives the three readings. Colder fields commit earlier in the reverse process: t_c falls from 1.61 at $\Theta = 0.2$ to 0.46 at $\Theta = 2$. The exact posterior commits at higher noise than its Gaussian twin at every temperature, by 0.47 in t when the field is coldest and 0.07 when it is hottest: at $\Theta = 0.2$ the exact posterior is half-committed when the noise variance is 24 times the signal's, its twin only at 8.9 times, so it is the two humps and not only the correlations that decide when the denoiser commits. At fixed $\Theta = 0.2$ the commitment time of the central sites grows with the chain length (1.03 at $L = 4$ to 1.72 at 64) and stops growing once the chain is several correlation lengths long (1.750 and 1.754 at 128 and 256); below the correlation length it moves one-for-one with $\frac{1}{2}\ln(1 + \lambda_{\max})$, $\lambda_{\max}$ the largest eigenvalue of the covariance (slope 0.98), the finite-chain analogue of the speciation time of Biroli et al. (2024). A second reading checks the lattice against the continuum: as $h \to 0$ the physical correlation length converges (it changes by 0.04% from $h = 1/16$ to $1/32$), and the dilute-kink law $\xi \propto \Theta^{1/2} e^{E_{\text{kink}}/\Theta}$ is approached as Θ falls but not reached on $\Theta \in [0.1, 0.3]$: a four-parameter fit returns a barrier of 0.988 against $2\sqrt{2}/3$ (a registered prediction, within 2%, failed), and we do not claim the asymptotic constant.

Beyond the score. The same fitted kernels serve other tasks without refitting (Appendix L). Hiding sites of a noised sequence and drawing posterior sequences by forward-filter backward-sample, the learned site energy and spline match the oracle within 0.4% in error and width and cover a nominal 90% interval 0.890–0.893 of the time, while on the base cell a Gaussian smoother with the true covariance is only 1.5%–4.6% worse in error, with 7%–9% wider intervals. On snapshots taken during coarsening after a quench, where the law is not Gibbs and the reference is therefore the clean snapshot (so a perfect denoiser does not reach zero and the ratios are compressed), the site energy still beats an empirical Gaussian smoother at every time (risk ratio 0.986 shortly after the quench, 0.950 at equilibrium), and the flexible spline adds nothing.

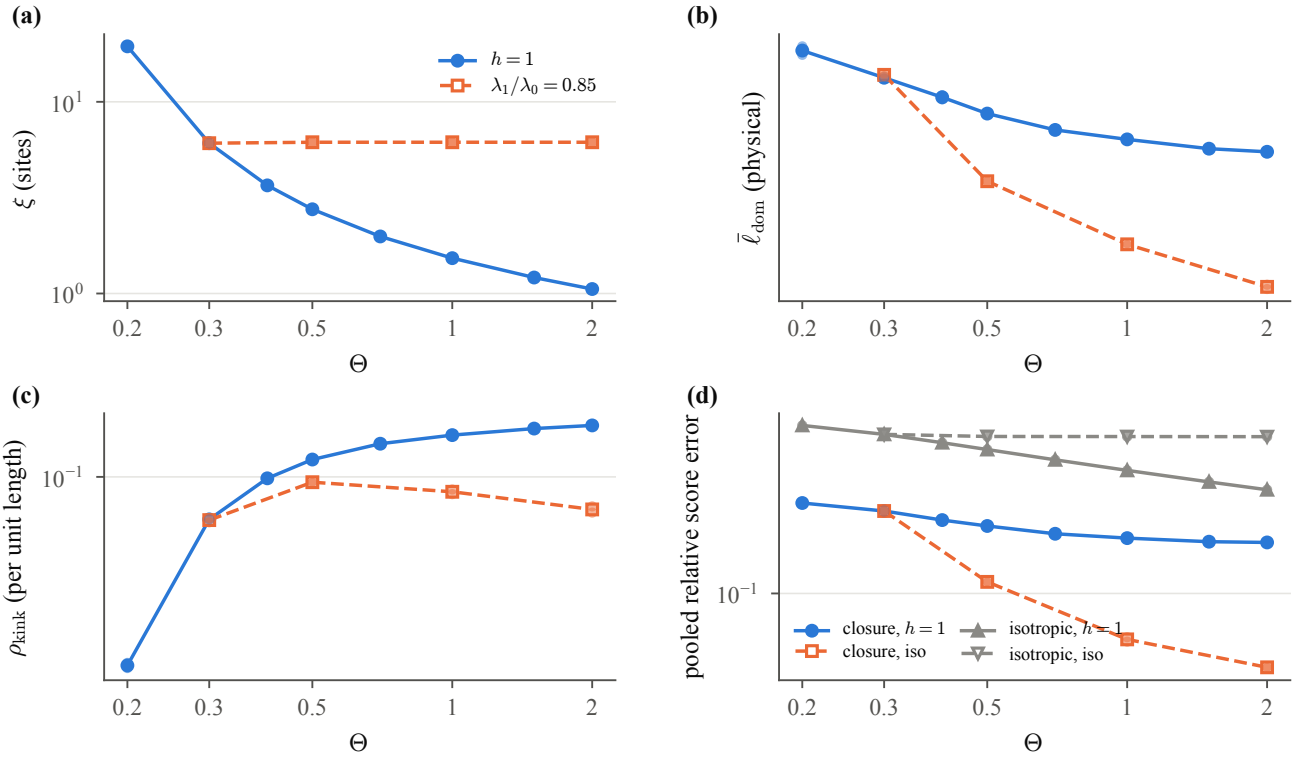


Figure 10. Temperature sweep of the field. (a) Correlation length in sites; (b) mean domain length (physical units); (c) coarse kink density per unit length; (d) pooled relative score error of the linear (Gaussian) closure and of the isotropic score, for the fixed-spacing axis $h = 1$ (blue) and the iso-correlation axis (orange), where h is solved so that the slowest correlation mode decays at the same rate at every Θ. The two axes share the base cell $\Theta = 0.3$.

8. Computation: what exact inference costs and how to cut it

The advantage of the structured estimator is in data and understanding, not in wall-clock, and the cost deserves its own measurement. Grid BP is the bottleneck of the method: each message update is a product of an $N_g \times N_g$ matrix with N_g-vectors, once per site and per level. We measure it on exclusive nodes (the whole node allocated), report the thread count with every number, and then ask whether the grid can be smaller or the messages represented differently.

Cost and its scaling. On the paper's 401-point grid with eight threads, one posterior mean of a 32-site sequence takes 0.52 ms against 0.042 ms for the largest transformer, a factor 12.5 on a CPU node and 15.1 on an H200 GPU (at batch 512; the GPU time is flat in the batch because BP is bound by kernel launches, so the ratio falls to 6.2 at batch 2048). The cost is linear in the number of sequences and in the chain length (log-log slopes 1.03 and 1.03) and grows like $N_g^{1.16}$ rather than N_g^2, because elementwise work and memory traffic dominate the matrix product at these sizes. Training is not cheaper than a network's: the E-step is linear in the number of sequences and the M-step does not depend on it, but total EM time is comparable with a network's, not orders of magnitude smaller (Appendix M). Memory is five message-sized arrays, 266 MB for a 512×32 batch at 401 points, linear in batch, length and grid, and 1.8–4.2 times the largest network.

The grid can be coarser. The 401-point grid was chosen for convergence safety and checked from above only (finer grids change nothing). Figure 13 checks it from below. With the true ϕ^4 kernel the score is exact to 1.8×10^{-11} at 101 points on every cell, because the trapezoid rule on an analytic integrand converges faster than any power of the spacing; and the learned ϕ^4 kernels lose nothing: at 101 points the site energy's and the spline's risks equal their 401-point values to three digits on two cells and at two sample sizes, and at 51 points they move by at most 1.2%, while the EM fit is 14 times faster. On one thread this makes BP 6.2 times faster at 101 points and 3.9 times at 151 than at 401. The linear chain's kernels are harder: with Laplace noise the transition is not smooth and, with many sequences, the eight-component mixture learns

Table 4. Risk divided by the linear floor (a Gaussian closure scores 1), mean over sixteen seeds, of every estimator on the six cells; the first four columns are the temperature sweep at matched correlation length, the last two refine the base cell at fixed physical length. EM's stopping point and each network's checkpoint are selected by the held-out risk against the exact score. A superscript counts failed fits, excluded from the mean; a dash means all sixteen failed. Bold: smallest in its column.

M	estimator	$\Theta = 0.3$	0.5	1	2	$h = 0.5$	$h = 0.25$
128	site energy (5)	0.139	0.269	0.534	0.630	0.136	0.130
	spline tilt (25)	0.218	0.454	1.111	1.478	0.231	0.235
	Legendre tilt (7)	0.394	0.354	0.647	0.816	0.330	0.387^{6}
	MDN	0.468	0.900	1.724	2.457	0.476	0.585
	mixture (paper)	1.018	1.039	1.187	1.279	1.016	1.017
	window, ISM	1.026	4.396	6.264	8.735	1.495	2.237
	transformer, ISM	3.284	9.752	8.509	14.487	3.591	4.729
	window, DSM	0.829	2.339	4.149	5.786	1.125	1.580
	transformer, DSM	1.787	4.359	8.034	11.571	1.887	2.129
2048	site energy (5)	0.033	0.072	0.141	0.169	0.040	0.032
	spline tilt (25)	0.060	0.134	0.280	0.429	0.073	0.068
	Legendre tilt (7)	0.074	0.092	0.165	0.243	0.111	–
	MDN	0.228	0.422	0.737	0.837	0.310	0.420
	mixture (paper)	1.004	1.004	1.014	1.021	1.010	1.016
	window, ISM	0.552	1.284	2.457	3.559	0.776	1.196
	transformer, ISM	1.354	2.706	2.440	3.237	1.368	1.020
	window, DSM	0.484	1.174	2.287	3.334	0.701	1.030
	transformer, DSM	0.533	1.225	2.156	3.043	0.622	0.780

components narrower than a coarse spacing, so its risk is within 0.08% of the 401-point value at 151 points but 2.2% off at 101 and 70% off at 51 ($M = 2048$), and only 22% of its kernels are resolved at 101 points (a registered prediction that at least 90% would be resolved failed). The safe grid is therefore about 101 points for the ϕ^4 kernels and about 151 for the mixtures, and the coarse grid cuts BP's cost by four to six times.

With one thread the comparison changes. The eight-thread ratio flatters the networks, which use threads, while BP barely does (0.667 ms on one thread against 0.52 on eight at 401 points). On one thread and the same node a network forward pass costs 0.023 ms (window), 0.063 (GRU), 0.152 (U-Net), 0.208 (transformer) and 0.216 (TCN), so BP at 401 points is 3.2 times slower than the transformer and 10.6 times slower than the GRU; at 151 points (0.173 ms) it is faster than the transformer and TCN, within 15% of the U-Net and 2.8 times slower than the GRU. The networks timed are the largest screened variants, several times the size of those scored. BP is also independent across sequences, so a node can run it as parallel processes; that was not measured. The honest summary is that exact inference costs more than a network forward pass by a factor that depends on the machine and the grid: 12.5 on a many-threaded CPU and 15.1 on a GPU at 401 points, and on one core between the cost of the largest networks and 2.8 times that of the fastest at 151 points. What it buys with that is an error 7.9 times smaller than the best network's at matched information on the linear chain.

Why not Fourier modes. BP's messages could be carried in Fourier space, where the convolution of the kernel with the likelihood is a product, truncated to a few modes. We measured the premise. The message does not have a short exact description: the natural parametric ansatz, a Gaussian passed once through the kernel, is exact at the first site only and not a fixed point of the recursion, so its error grows into the chain by orders of magnitude (it fails to close in all 432 configurations tested), while it is 2,272 times closer to the true message than a Gaussian at low noise. As a representation a Fourier basis fitted to each message needs 33 numbers for a KL divergence of 10^{-2} against 45 for the grid, and 47 for 10^{-4} against 151 (on the linear chain with Laplace noise), but it reaches 10^{-4} in only 53 of 144 of the cases and adding modes can make it worse, so a solver cannot pick a truncation and know its error; the cumulant (Gram–Charlier) route is non-monotone in most cases. As a solver it would help little: the step Fourier speeds up, the transition matrix product, is only 12.1%–15.2% of a BP sweep at 401 points, capping any speed-up at 1.14–1.18×, and the FFT transition we built is second-order accurate (6.7×10^{-4} from dense BP) but 5.8 times slower at 401 points, and still 2.2 times slower at the largest grid we timed (3201 points). Coarsening reduces every term

Table 5. Samples drawn by one forward sweep from each fitted kernel at $M = 2048$, against exact samples of the same cell (four seeds): coarse kink density and coarse mean domain length, and the 10th, 50th and 90th percentiles of the domain-length distribution (kinks, domain, q_{10}, q_{50}, q_{90}), each as generated over exact (mean over seeds of the per-seed ratio; 1 is exact), and the Kolmogorov–Smirnov distance between the two domain-length distributions (KS).

		$\Theta = 0.3$	0.5	1	2	$h = 0.5$	$h = 0.25$
site energy	kinks	0.987	0.993	0.999	1.004	1.012	0.979
	domain	1.016	1.013	0.999	1.013	0.995	0.990
	q_{10}	1.000	1.000	1.036	1.000	1.042	0.944
	q_{50}	1.000	1.028	1.021	1.019	0.969	0.991
	q_{90}	1.000	1.000	0.983	1.021	1.000	0.999
	KS	0.018	0.020	0.053	0.073	0.019	0.025
spline tilt	kinks	0.989	1.009	1.009	0.999	0.999	0.995
	domain	1.016	1.018	1.004	1.000	0.999	0.994
	q_{10}	1.000	1.000	1.036	1.000	1.104	0.972
	q_{50}	1.000	1.028	1.001	1.003	1.011	0.987
	q_{90}	1.017	1.000	1.004	1.001	1.008	1.004
	KS	0.022	0.022	0.035	0.065	0.015	0.021
mixture (paper)	kinks	1.405	1.095	1.021	0.911	1.425	1.405
	domain	0.873	1.002	1.008	1.039	0.862	0.878
	q_{10}	1.000	1.000	1.004	1.000	0.875	0.944
	q_{50}	0.833	0.975	1.049	1.063	0.821	0.857
	q_{90}	0.878	1.000	0.993	1.041	0.874	0.885
	KS	0.081	0.021	0.062	0.069	0.085	0.079

of the sweep, which is why it is the better lever. Appendix M gives the tables.

9. Where the structure stops

The structured estimator assumes four things: that the data are Markov, that the kernel family is wide enough to contain the truth, that the grid resolves the kernel, and that the law is stationary. Each can fail, and each failure has a signature we can measure.

9.1. The kernel family: when does a flexible kernel pay?

The simple kernels are exactly specified on the two chains above, so the question is what happens when they are not. Two experiments take the truth out of the family, and one asks the neural kernel the same question.

An unrestricted kernel on the field. On the ϕ^4 field the 100-parameter unrestricted kernel, which assumes no Gaussian bond, costs 2.5–5.5 times the site energy's risk, and the price grows with the data rather than shrinking (2.8 at $M = 128$ to 5.5 at 2048 on the base cell): the exactly specified model improves roughly like the sample size and the unrestricted one more slowly. Generality is paid for where it is not needed.

A chain outside the family. Replace the Gaussian bond by a sharp Laplace bond: the pair energy $|a' - a|$ in place of $(a' - a)^2$ on the same double well, with the bond scale and the well depth calibrated beforehand to the base cell's lag-one correlation and kurtosis (Appendix L). This chain is outside the tilted-walk class, so the site energy and the spline are misspecified; its exact samples and oracle are available from the grid. The site energy and the spline stall at a risk of about 0.10 while the unrestricted kernel keeps improving, overtakes them between $M = 128$ and 512, and ends 1.5× below the site energy at 2048 in every one of eight seeds. A neural kernel does better still: the MDN reaches 0.056 at $M = 2048$ against 0.068 for the unrestricted kernel and 0.103 for the site energy, below the site energy in 7 of eight seeds and, against a registered prediction that parsimony would win, below the unrestricted kernel in 7 (its fits were still improving at the last checkpoint in 5 of eight, so its risk is an upper bound).

Subsampling the field. Observing a fine lattice only at every s-th site gives a chain that is still exactly Markov, with kernel K^s, but no longer a tilted walk. The coarse-grained physics is unchanged across strides (kink density and domain length agree within 7.6%), but the five- and twenty-five-parameter kernels are now misspecified: their risk at stride 4 is 25 and 14 times its stride-one value (kernels fitted on clean chains, sixteen seeds), though it stays far below the linear floor (0.040 and 0.036 against 0.235). Here the network kernel earns its parameters: with its ladder extended to 16 000 updates the MDN is below the site energy at

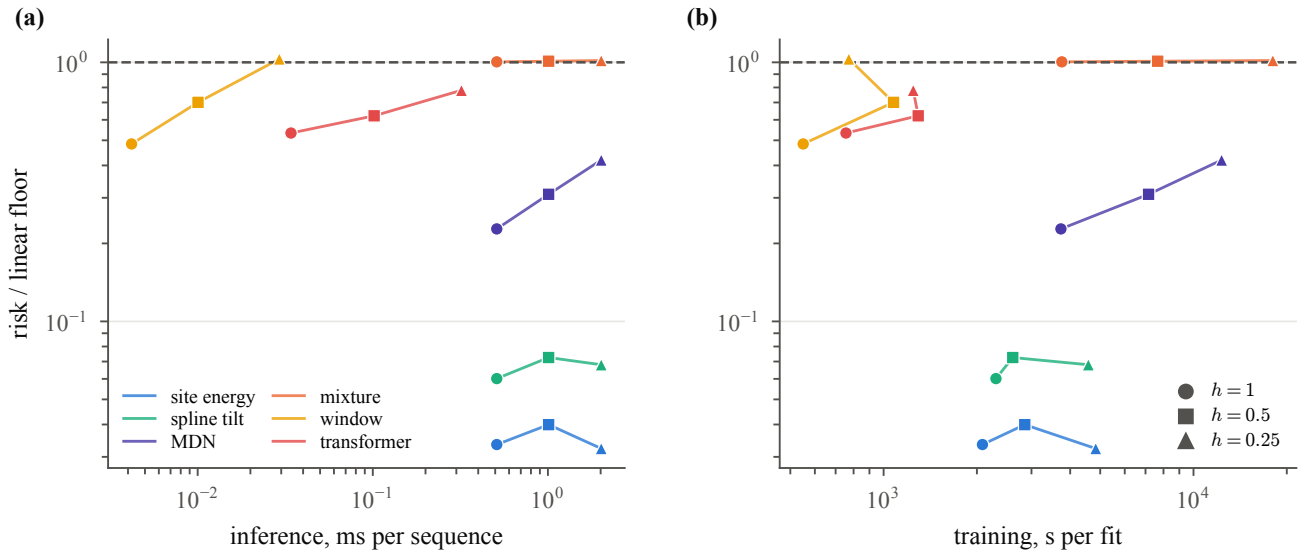


Figure 11. Accuracy against cost on the field. Risk divided by the linear floor against (a) inference time per sequence ($M = 2048$ training sequences; eight threads, h-matched cells, $B = 512$, the largest network variants, so the networks' times are upper bounds) and (b) training time per fit (shared nodes, indicative). Circle, square and triangle: $h = 1, 0.5, 0.25$ at $\Theta = 0.3$ with 32, 64, 128 sites. The dashed line is the linear floor.

stride 4 in 16 of sixteen seeds at 0.40 times its risk, while at stride 1, where the site energy is exactly specified, the site energy is 3.0 times better. The MDN was still improving at its last checkpoint in 11 of the sixteen fits, so its risks are upper bounds and the margin can only widen.

The rule that emerges is simple. A kernel with a few parameters that contains the physics is better than a flexible one, the paper's own linear-drift mixture is correct for a linear chain and wrong for a double well, a network kernel loses to it on the linear chain by a factor 2.1–2.9 and also to the best plain score network, and the flexible kernels (unrestricted, neural) are the right tools exactly when the simple family is wrong. The unrestricted kernel is therefore best used as a misspecification test: if it beats the family it is compared with at large M, the family's assumption is wrong.

9.2. The Markov assumption: a latent shared by the whole sequence

The estimator is a model of $p(a_{i+1} \mid a_i)$ only. If a hidden variable is shared by a whole sequence, the observed chain is no longer Markov, a bias remains that more data cannot remove, and its size is computable. We start with a closed-form case on the linear chain: $X_i = Gb + U_i$ with $G = \pm 1$ a fair sign per sequence and U the Gaussian linear chain, whose exact score is a mixture of two Gaussian chains. The first-order kernel then stops improving: at $b = 2$ four times more data change its risk by +0.2%, and it ends 1.56–1.71 times worse than the linear floor in every one of sixteen seeds. The repair is the same mixture kernel on the augmented state (X_i, G), the two phases tied by reflection, with the same number of parameters: BP runs on pairs, inferring each sequence's sign, and the risk keeps falling with the data (52% over the same factor of four), ending 6.3 times below the linear floor and 13.1 times below the first-order kernel at $b = 2$ (Figure 14). A surprise is instructive: the repaired model also helps when there is no latent ($b = 0$). We traced it to the reflection symmetry of the data: fitting the plain kernel on each sequence together with its sign-flipped copy is 32%–39% better than the plain fit, and in average risk it is worth as much as fitting on twice as many sequences. The same mechanism for general departures from Markov dependence is measured in Figure 15.

Gaussian departures from the Markov structure. Every departure is Gaussian so that its score is exact, and every arm is scored against that score; the network arm is a radius-2 window head under the recipe of Appendix C. The ordering degrades smoothly in each mechanism and the strength at which it breaks even is a property of the mechanism, not a single number. A rank-one global latent never reverses the ordering in the swept range: the ratio falls only from 15.55 to 3.97 across the swept range, up to $\delta = 1$. A second-lag coefficient reverses it between 0.3 and 0.4 at both sizes, and a site-dependent ρ between 0.1 and 0.14 at $M = 2048$. Long-range precision coupling does not re-

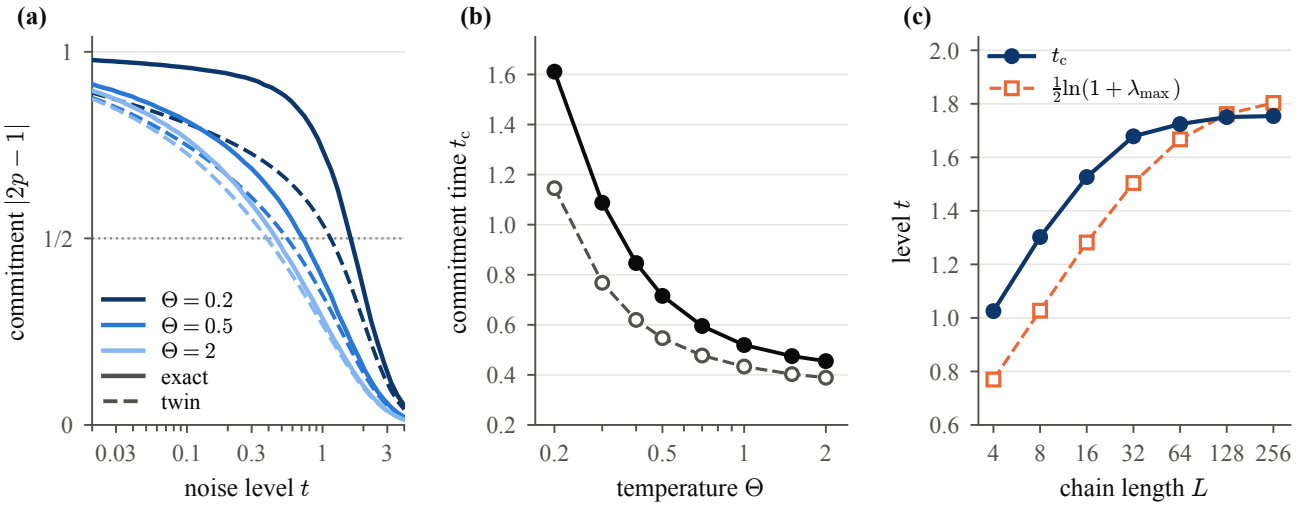


Figure 12. Commitment of the exact denoiser. (a) Site-averaged commitment $|2p-1|$ against the noise level t at three temperatures, for the exact posterior (solid) and the Gaussian twin (dashed). (b) The level t_c where the commitment crosses one half against Θ, exact (filled) and twin (open), at the fixed spacing $h = 1$, $L = 32$. (c) At $\Theta = 0.2$, t_c of the central sites against the chain length L (filled) and $\frac{1}{2}\ln(1+\lambda_{\max})$ (open).

verse it anywhere we swept: the ratio decays to 1.00 at $\delta = 0.4$, a tie rather than a crossing. We report brackets and not fitted crossings, because with 16 seeds the paired log-ratio is resolved to a few thousandths and the crossing sits between grid points either way.

The two sizes separate the statistical gain of the structure at small M from its asymptotic bias at large M, and the bias has a closed form: every truth is Gaussian and the fitted driving noise is Gaussian to three decimals, so the $M \to \infty$ limit of EM–BP is the Kullback–Leibler projection of the noised truth onto the linear-Gaussian family, whose relative score error $b(\delta)$ is a trace formula (Appendix K). It reproduces the measured EM–BP error at $M = 2048$ to within 0.019 across every mechanism and strength, $\sqrt{b(\delta)^2 + v/M}$ predicts both sizes (Figure 15, bottom row), and its order in the strength says which mechanisms bite: $b \propto \delta$ for the three mechanisms that perturb of order L directions of the covariance and $b \propto \delta^2$ for the rank-one latent, which perturbs one. Past the break-even the chain estimator has converged to a wrong law and more data buys nothing back, exactly the behaviour Section 6.1 attributes to a model whose bias, not whose variance, is binding.

9.3. Resolution, discretisation and what we do not claim

A flexible kernel can need a finer grid than the one the simple kernels use (Section 8), and the lattice itself is a modelling choice: the correlation length in sites grows like $1/h$, so a fixed-window network falls behind under refinement (Section 7) and the advantage of structure grows. We also do not claim the following. Superiority over networks in general: the comparison equalises the number of sequences under each method's standard recipe and, separately, matches the information interface with no oracle on either side; it does not rank learning algorithms, and the networks timed in Section 8 are the largest variants. Exactness beyond quadrature: grid BP is exact for the discretised model and accurate to the measured quadrature, and we say which where it matters. A Kalman filter: the Gaussian pass is a smoother. A phase transition: the field's low-temperature crossover is smooth in one dimension. A Laplace closure: the transition-family message ansatz is not exact beyond the first site. A general SPDE transition: a fixed driving-noise family such as the mixture kernel is a different object from the field's tilted walk, and it is misspecified there through its drift.

10. Related work

Solvable diffusion and the Gaussian part of the score. Diffusion and score matching are due to Sohl-Dickstein et al. (2015); Ho et al. (2020); Song et al. (2021) and Hyvärinen (2005); Vincent (2011); the score–posterior-mean identity is Tweedie's (Robbins, 1956; Efron, 2011). Exactly solvable targets have been Gaussian (Pierret & Galerne, 2024), mean-field or mixture-like (Biroli & Mézard, 2023; Biroli et al., 2024; Cui et al., 2025), or hierarchical with discrete leaves (Sclocchi et al., 2025), on which U-Nets and transformers imple-

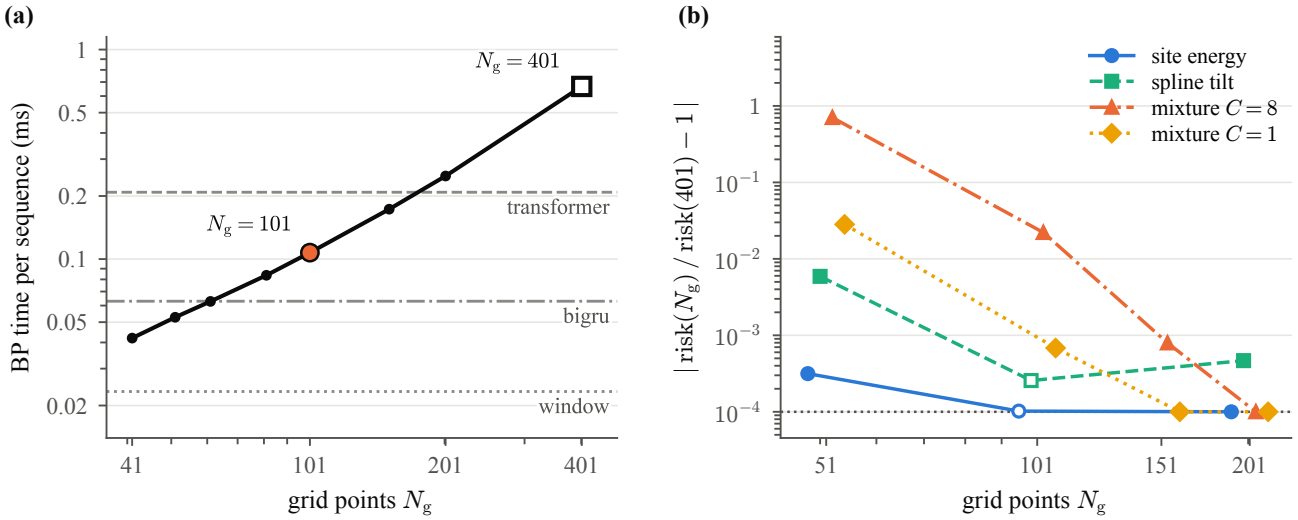


Figure 13. A coarser grid. (a) BP time per sequence against the grid size N_g (one thread, an exclusive node, $B = 512$, $L = 32$), with the times of three networks on the same node. (b) Relative change of the risk against its 401-point value (the absolute value of the seed-mean ratio minus one) for the ϕ^4 site energy and spline (base cell, $M = 2048$; no fit at 151 points) and for the linear chain's mixture kernels with $C = 1$ and $C = 8$ (frozen kernels, $M = 2048$). Hollow markers: the coarse fit is better than the 401-point fit; points within 10^{-4} of it are drawn at that floor (dotted); the four series are offset horizontally so that coincident points can be told apart.

ment belief propagation (Mei, 2024; Mei & Wu, 2023; Garnier-Brun et al., 2024). A recurrent empirical observation is that learned scores are dominated by their Gaussian part over a wide noise range (Wang & Vastola, 2024; Li et al., 2024; Kadkhodaie et al., 2024; Wang et al., 2026), and that downstream statistics depend on the first two moments alone (Ghane et al., 2025). Our chain is the degenerate tree of depth L with continuous non-Gaussian states; the Gaussian part of its score is the Kalman smoother of the covariance twin (Proposition 1), and Equation (7) is the sharp rate, with its constant and crossover, behind the dominance those papers report.

What survives noise: the low-SNR literature. Theorem 5 belongs to the low-SNR expansions of estimation theory. In Gaussian channels the mutual information and the MMSE expand in the SNR with fourth-order statistics entering at second order, for arbitrary input laws (Prelov & Verdú, 2004; Guo et al., 2005; Dytso et al., 2023; Kong et al., 2023); for latent-variable models the log-likelihood expands with the order-k term matching k-th moments (Katsevich & Bandeira, 2023), which yields the σ^{2k} sample complexity of multireference alignment and orbit recovery (Perry et al., 2019; Bandeira et al., 2023) and, most closely, the layered Fisher-information geometry of Balanov et al. (2026), where information about directions first identifiable at moment order k scales as SNR^k and the method of moments is almost efficient. Under the OU channel $\mathrm{SNR}_t \simeq e^{-2t}$, so that statement and our e^{-2mt} are the same law in different clocks. Aarts et al. (2025) write the forward-process cumulant attenuation e^{-nt} and show that score networks regenerate the cumulants. The information law is therefore not new here; what this paper adds is the score-level statement with its constant $\frac{1}{6}\|\kappa_4\|_F^2$ and its content (the fourth-order term closes the gap), the general-order constant, the hypothesis under which the expansion is a theorem and the heavy-tailed regime outside it, the crossover t_* as an exact scale, the verification of rates and constants against an exact non-Gaussian dependent score, and the learning consequences with an exact E-step. Speciation and collapse times (Biroli et al., 2024; Achilli et al., 2026) and the information-restricted Bayesian analysis of Hunt et al. (2026) are second-moment phenomena in this vocabulary; heavy-tailed forward processes (Pandey et al., 2024; Tang et al., 2026) change the channel and therefore the ladder.

Learning from noisy data only. Ambient diffusion learns a clean distribution from corrupted samples by predicting a further corruption (Daras et al., 2023; 2024); EM with diffusion posterior sampling in the E-step (Rozet et al., 2024; Bai et al., 2024) and SURE-type surrogates (Kawar et al., 2024) address the same problem nonparametrically. Our learner is parametric and its E-step is exact, so the question changes from "can the clean law be recovered" to "how fast is it lost", which Theorem 5 answers order by order. The estima-

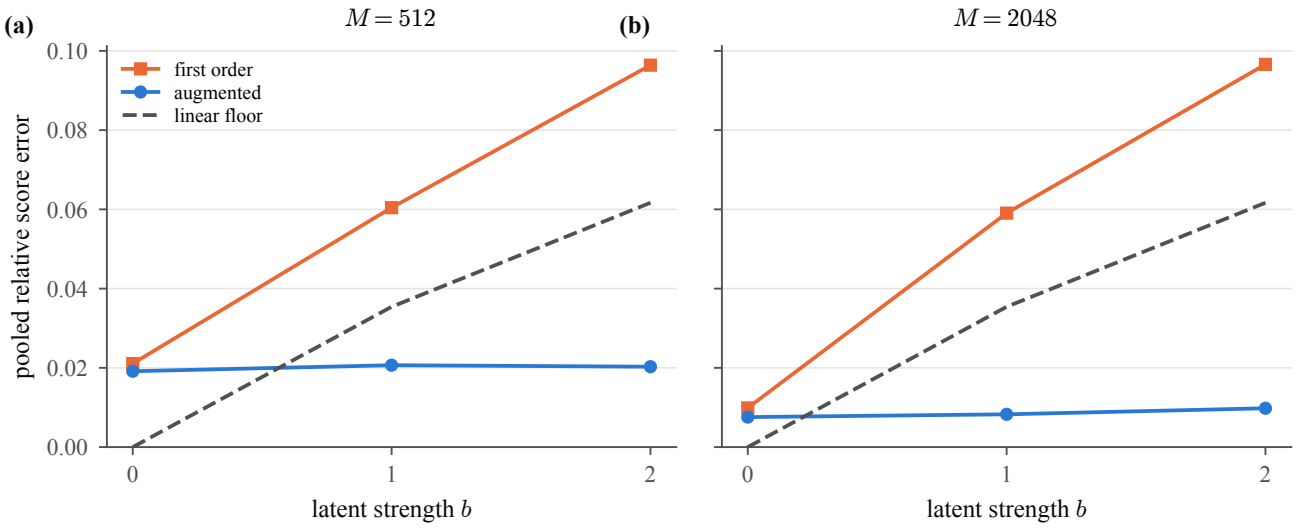


Figure 14. A global latent and its repair. Pooled relative score error against the latent strength b for the first-order kernel and for the same kernel on the augmented state (X_i, G), against the linear floor (dashed), at $M = 512$ (a) and 2048 (b); sixteen seeds. The first-order kernel's error does not fall with the data and ends above the linear floor; the augmented kernel's does, and ends far below it.

tor is classical: sum-product on a chain is forward–backward (Baum et al., 1970; Rabiner, 1989), grid quadrature is Kitagawa's smoother (Kitagawa, 1987), Fisher's identity and EM are Dempster et al. (1977); Cappé et al. (2005), and the Gaussian special case is the Shumway–Stoffer algorithm (Shumway & Stoffer, 1982). Consistency of the maximum-likelihood estimator for general-state hidden Markov models is Douc et al. (2011); the identifiability of a Markov signal observed through independent noise, which is what makes the one-view protocol of Section 5 well posed, is Gassiat et al. (2020). Spectral and moment methods for hidden Markov models (Hsu et al., 2012) are the natural high-noise competitor; the moment estimators of Section 5 are their simplest form here, rate-correct and inefficient below the crossover.

Physical Markov models and the field benchmark. A nearest-neighbour energy on a line is the one-dimensional Gibbs measure of statistical mechanics, solved exactly by a transfer matrix (Ising, 1925), which is also why it has no phase transition at any positive temperature; its Markov structure on a lattice is the Hammersley–Clifford picture of Besag (1974). Our ϕ^4 benchmark is a one-dimensional instance, chosen so that the exact score, the exact samples and the exact physical statistics are available from one eigensolve and every estimator can be judged against them and against a physical reading (temperature, correlation length, kink density). The same section of the literature that studies when generative diffusion commits to a mode, speciation (Biroli et al., 2024; Achilli et al., 2026), is the vocabulary of Section 7's commitment measurement, which we make exact on a chain; and the cumulant attenuation of the forward process (Aarts et al., 2025) is the same law as Theorem 3.

Sequence denoisers and length. The baselines of Section 6 are the standard sequence architectures (Bai et al., 2018; Vaswani et al., 2017; Ronneberger et al., 2015); that attention-based length extrapolation turns on the positional encoding (Press et al., 2022; Kazemnejad et al., 2023) is visible in Table 9, here against an exact reference at every length.

11. Conclusion

For data that form a Markov chain, score estimation under diffusion noise reduces to posterior means, and a forward–backward sweep of belief propagation computes them exactly for any continuous kernel, up to a quadrature we measure. The kernel itself is learnable from noised sequences alone, by expectation–maximisation around the same sweep, so one can learn the local law and infer globally. The Gaussian kernel is a clean check, since BP with Gaussian messages is the linear score and the Kalman smoother. On two chains whose scores are known, a linear chain with non-Gaussian noise and the one-dimensional ϕ^4 field, this estimator, with a kernel of a few parameters, beats the neural denoisers it is compared with at every sample size, including at its own information interface; the kernel family is the real modelling choice, the five-parameter site energy being right for a gradient flow

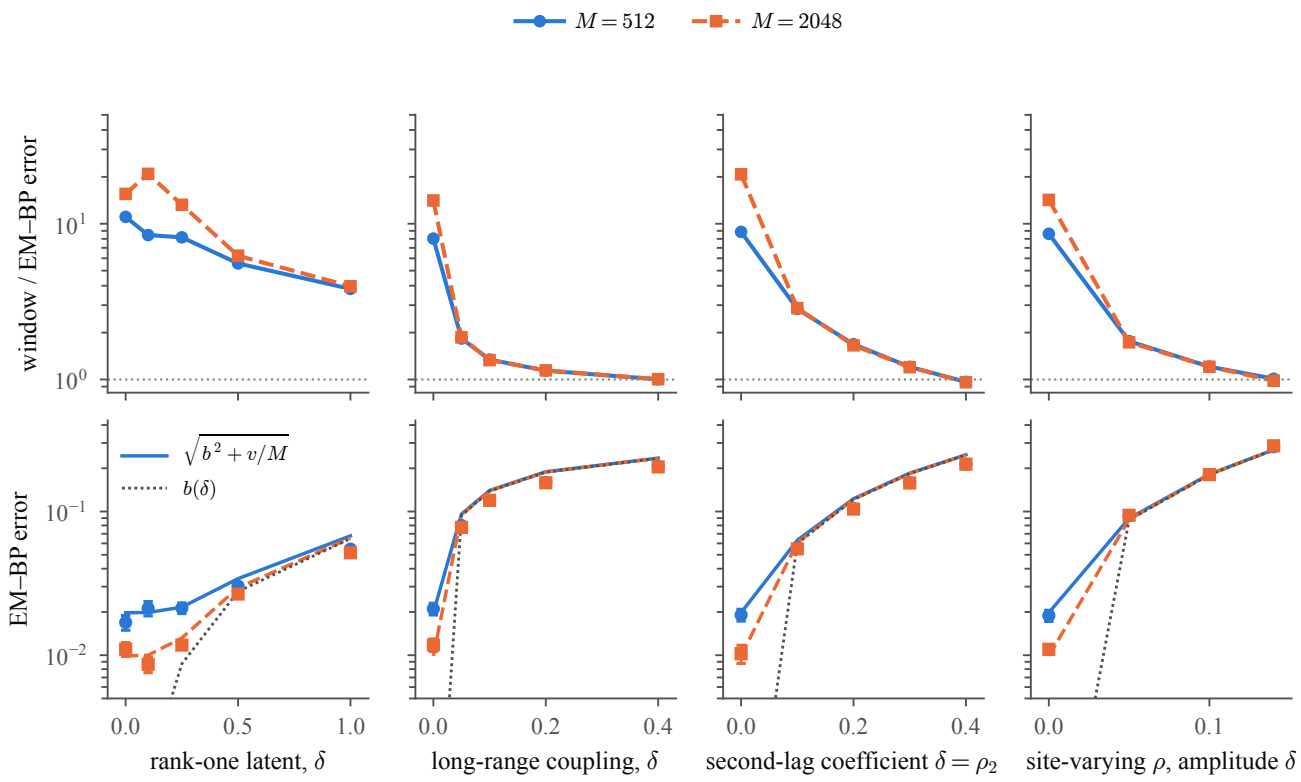


Figure 15. Where the structure stops helping. Ratio of a radius-2 window head's error to EM–BP's against the misspecification strength δ, for four Gaussian departures from the chain with exact reference scores (rank-one global latent, long-range precision coupling, a second-lag coefficient at fixed lag-one correlation, site-dependent ρ), at two sample sizes (top row). Above the dotted line the misspecified chain estimator still wins. Bottom row: EM–BP's own error at both sizes (points, paired standard errors) against the closed-form bias $b(\delta)$ of the Kullback–Leibler projection (dotted) and the finite-sample prediction $\sqrt{b(\delta)^2 + v/M}$ (lines) with v read from the well-specified cells (Appendix K).

and the paper's linear-drift mixture right for a linear chain and wrong for a double well; and a neural or unrestricted kernel earns its parameters only where the simple family is misspecified.

The advantage is in data and understanding, not in wall-clock. Exact inference costs more per sequence than a network forward pass, by a factor that depends on the machine and the grid, a coarser grid and a smaller kernel recover a large part of it for the field kernels, and a Fourier representation of the messages is not worth building. The structure also has limits that can be computed: a latent shared by a whole sequence leaves a bias no sample size removes, which augmenting the state repairs; a kernel outside the family leaves a floor the unrestricted kernel is built to detect; a coarse grid fails for narrow kernels. The comparison equalises the number of sequences under each method's standard recipe and, separately, matches the information interface with no oracle on either side; it does not rank learning algorithms in general, and the networks we time are the largest variants.

What the linear laboratory adds is a general fact about the channel: diffusion noise erases statistical structure order by order, at rates fixed by Hermite degree, so a score learner that reaches only second order is bounded away from the truth by an e^{-8t}-scaled term whose content is the fourth-order moment. This explains where the structured estimator gains and does not enter its exactness. Explicit structure recovers what the data still carry, and the price of the wrong structure is a bias that can be computed.

Impact statement

This paper presents theoretical and empirical work on how diffusion noise erases statistical structure and on learning structured dynamics from noised data, evaluated on synthetic models with known ground truth. Its societal consequences are those of a better understanding of generative diffusion models in general; we see no specific ethical concern that requires discussion here.

References

Aarts, G., Habibi, D. E., Wang, L., and Zhou, K. On learning higher-order cumulants in diffusion models. Machine Learning: Science and Technology, 6: 025004, 2025. arXiv:2410.21212.

Achilli, B., Benedetti, M., Biroli, G., and Mézard, M. Theory of speciation transitions in diffusion models with general class structure, 2026.

Bai, S., Kolter, J. Z., and Koltun, V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling, 2018. arXiv:1803.01271.

Bai, W., Wang, Y., Chen, W., and Sun, H. An expectation-maximization algorithm for training clean diffusion models from corrupted observations. In Advances in Neural Information Processing Systems, 2024. arXiv:2407.01014.

Balanov, A., Bendory, T., and Edidin, D. The generalized method of moments is (almost) statistically efficient in low-SNR Gaussian latent-variable models, 2026. arXiv:2605.30095.

Bandeira, A. S., Blum-Smith, B., Kileel, J., Perry, A., Niles-Weed, J., and Wein, A. S. Estimation under group actions: recovering orbits from invariants. Applied and Computational Harmonic Analysis, 66: 236–319, 2023.

Baum, L. E., Petrie, T., Soules, G., and Weiss, N. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. The Annals of Mathematical Statistics, 41(1):164–171, 1970.

Besag, J. Spatial interaction and the statistical analysis of lattice systems. Journal of the Royal Statistical Society: Series B, 36(2):192–225, 1974.

Biroli, G. and Mézard, M. Generative diffusion in very large dimensions. Journal of Statistical Mechanics: Theory and Experiment, 2023(9):093402, 2023.

Biroli, G., Bonnaire, T., de Bortoli, V., and Mézard, M. Dynamical regimes of diffusion models. Nature Communications, 15:9957, 2024. doi: 10.1038/s41467-024-54281-3.

Cappé, O., Moulines, E., and Rydén, T. Inference in Hidden Markov Models. Springer, 2005.

Cui, H., Pehlevan, C., and Lu, Y. M. A solvable model of learning generative diffusion: theory and insights, 2025.

Daras, G., Shah, K., Dagan, Y., Gollakota, A., Dimakis, A. G., and Klivans, A. Ambient diffusion: learning clean distributions from corrupted data. In Advances in Neural Information Processing Systems, 2023. arXiv:2305.19256.

Daras, G., Dimakis, A. G., and Daskalakis, C. Consistent diffusion meets Tweedie: training exact ambient diffusion models with noisy data. In International Conference on Machine Learning, 2024. arXiv:2404.10177.

Dempster, A. P., Laird, N. M., and Rubin, D. B. Maximum likelihood from incomplete data via the EM algorithm. Journal of the Royal Statistical Society, Series B, 39(1):1–38, 1977.

Douc, R., Moulines, E., Olsson, J., and van Handel, R. Consistency of the maximum likelihood estimator for general hidden Markov models. Annals of Statistics, 39(1):474–513, 2011.

Dytso, A., Poor, H. V., and Shamai, S. Conditional mean estimation in Gaussian noise: a meta derivative identity with applications. IEEE Transactions on Information Theory, 69:1883–1910, 2023.

Efron, B. Tweedie's formula and selection bias. Journal of the American Statistical Association, 106 (496):1602–1614, 2011.

Garnier-Brun, J., Mézard, M., Moscato, E., and Saglietti, L. How transformers learn structured data: insights from hierarchical filtering, 2024.

Gassiat, E., Le Corff, S., and Lehéricy, L. Identifiability and consistent estimation of nonparametric translation hidden Markov models with general state space. Journal of Machine Learning Research, 21(115):1–40, 2020.

Ghane, R., Bao, A., Akhtiamov, D., and Hassibi, B. Gaussian universality for diffusion models, 2025. arXiv:2501.07741.

Guo, D., Shamai, S., and Verdú, S. Mutual information and minimum mean-square error in Gaussian channels. IEEE Transactions on Information Theory, 51(4):1261–1282, 2005. doi: 10.1109/TIT.2005.844072.

Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In Advances in Neural Information Processing Systems (NeurIPS), volume 33, pp. 6840–6851, 2020.

Hsu, D., Kakade, S. M., and Zhang, T. A spectral algorithm for learning hidden Markov models. Journal

of Computer and System Sciences, 78(5):1460–1480, 2012.

Hunt, H., Kamb, M., and Ganguli, S. An exact information theory of generalization phase transitions in Bayesian diffusion models, 2026. arXiv:2607.08041.

Hyvärinen, A. Estimation of non-normalized statistical models by score matching. Journal of Machine Learning Research, 6:695–709, 2005.

Ising, E. Beitrag zur Theorie des Ferromagnetismus. Zeitschrift für Physik, 31(1):253–258, 1925.

Kadkhodaie, Z., Guth, F., Simoncelli, E. P., and Mallat, S. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In International Conference on Learning Representations, 2024. Outstanding Paper award.

Katsevich, A. and Bandeira, A. S. Likelihood maximization and moment matching in low SNR Gaussian mixture models. Communications on Pure and Applied Mathematics, 76(4):788–842, 2023. arXiv:2006.15202.

Kawar, B., Elata, N., Michaeli, T., and Elad, M. GSURE-based diffusion model training with corrupted data. Transactions on Machine Learning Research, 2024. arXiv:2305.13128.

Kazemnejad, A., Padhi, I., Natesan Ramamurthy, K., Das, P., and Reddy, S. The impact of positional encoding on length generalization in transformers. In Advances in Neural Information Processing Systems, 2023. arXiv:2305.19466.

Kitagawa, G. Non-Gaussian state-space modeling of nonstationary time series. Journal of the American Statistical Association, 82(400):1032–1041, 1987.

Kong, X., Brekelmans, R., and Ver Steeg, G. Information-theoretic diffusion. In International Conference on Learning Representations, 2023. arXiv:2302.03792.

Kschischang, F. R., Frey, B. J., and Loeliger, H.-A. Factor graphs and the sum–product algorithm. IEEE Transactions on Information Theory, 47(2): 498–519, 2001.

Li, X., Dai, Y., and Qu, Q. Understanding generalizability of diffusion models requires rethinking the hidden Gaussian structure. In Advances in Neural Information Processing Systems, 2024. arXiv:2410.24060.

Mei, S. U-nets as belief propagation: Efficient classification, denoising, and diffusion in generative hierarchical models, 2024. arXiv:2404.18444.

Mei, S. and Wu, Y. Deep networks as denoising algorithms: sample-efficient learning of diffusion models in high-dimensional graphical models, 2023. arXiv:2309.11420; IEEE Transactions on Information Theory, Xplore document 10857317.

Meng, X.-L. and Rubin, D. B. Maximum likelihood estimation via the ECM algorithm: A general framework. Biometrika, 80(2):267–278, 1993.

Pandey, K., Pathak, J., Xu, Y., Mandt, S., Pritchard, M., Vahdat, A., and Mardani, M. Heavy-tailed diffusion models, 2024. arXiv:2410.14171.

Pearl, J. Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference. Morgan Kaufmann, San Mateo, CA, 1988.

Perry, A., Weed, J., Bandeira, A. S., Rigollet, P., and Singer, A. The sample complexity of multireference alignment. SIAM Journal on Mathematics of Data Science, 1(3):497–517, 2019.

Pierret, E. and Galerne, B. Diffusion models for Gaussian distributions: exact solutions and Wasserstein errors, 2024. arXiv:2405.14250.

Prelov, V. V. and Verdú, S. Second-order asymptotics of mutual information. IEEE Transactions on Information Theory, 50(8):1567–1580, 2004.

Press, O., Smith, N. A., and Lewis, M. Train short, test long: attention with linear biases enables input length extrapolation. In International Conference on Learning Representations, 2022. arXiv:2108.12409.

Rabiner, L. R. A tutorial on hidden Markov models and selected applications in speech recognition. Proceedings of the IEEE, 77(2):257–286, 1989.

Robbins, H. An empirical Bayes approach to statistics. In Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, volume 1, pp. 157–163, 1956.

Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation. In Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 234–241, 2015.

Rozet, F., Andry, G., Lanusse, F., and Louppe, G. Learning diffusion priors from observations by expectation maximization. In Advances

in Neural Information Processing Systems, 2024. arXiv:2405.13712.

Sclocchi, A., Favero, A., and Wyart, M. A phase transition in diffusion models reveals the hierarchical nature of data. Proceedings of the National Academy of Sciences, 122(1):e2408799121, 2025.

Shumway, R. H. and Stoffer, D. S. An approach to time series smoothing and forecasting using the EM algorithm. Journal of Time Series Analysis, 3(4): 253–264, 1982.

Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., and Ganguli, S. Deep unsupervised learning using nonequilibrium thermodynamics. In Proceedings of the 32nd International Conference on Machine Learning (ICML), pp. 2256–2265, 2015.

Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. Score-based generative modeling through stochastic differential equations. In International Conference on Learning Representations (ICLR), 2021.

Tang, W., Touzi, N., Zhang, Z., and Zhou, X. Y. Tweedie's formulae and diffusion generative models beyond Gaussian, 2026. arXiv:2605.19391.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In Advances in Neural Information Processing Systems (NeurIPS), volume 30, 2017.

Vincent, P. A connection between score matching and denoising autoencoders. Neural Computation, 23(7): 1661–1674, 2011.

Wang, B. and Vastola, J. J. The unreasonable effectiveness of Gaussian score approximation for diffusion models and its applications. Transactions on Machine Learning Research, 2024. arXiv:2412.09726.

Wang, B., Zavatone-Veth, J., and Pehlevan, C. A random matrix theory perspective on the consistency of diffusion models. In International Conference on Machine Learning, 2026. arXiv:2602.02908.

A. Proofs

A.1. Proposition 1

Write $\overrightarrow{m}_i^{\mathrm{p}}$ for the forward message at site i after projection and ℓ_i for the Gaussian unary likelihood. We show by induction that every projected message equals the corresponding exact message of the covariance-matched Gaussian chain. The initial message is Gaussian or projected to the Gaussian with the same first two moments, which is the Gaussian chain's initial message. If $\overrightarrow{m}_{i-1}^{\mathrm{p}}$ is that chain's message, with mean m and variance v, multiplying by ℓ_{i-1} gives a Gaussian with the precision-weighted moments, identical in both chains; the transition integral against $a_i = \rho a_{i-1} + \varepsilon_i$ is the density of $\rho a_{i-1} + \varepsilon_i$, whose mean and variance are $(\rho m', \rho^2 v' + q)$ for every driving noise with moments $(0, q)$, so Π_{G} returns the Gaussian chain's next message. The backward message is $h(\rho a)$ with h the density of $a'' - \varepsilon$ under the incoming Gaussian, of mean m' and variance $v' + q$; as a function of a its normalised moments are $(m'/\rho, (v'+q)/\rho^2)$, which is where $\rho \neq 0$ is used, starting from the terminal likelihood ℓ_L (the flat terminal message is not normalisable and contributes no factor). Every projected belief is therefore the Gaussian chain's belief, whose posterior mean is linear in $\mathbf{x}$ and coincides with the LMMSE estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Since the recursion never read p_ε beyond $(0, q)$, the estimator is the same for every such law; nothing assumes stationarity, since the induction carries site-dependent moments. The algebraic steps (the two moment maps as integrals, the covariance identities, and the optimality of the normal-equation gain in arbitrary dimension) are machine-checked in Lean 4 (lean/README.md); the projection step itself is not. The two consequences stated after the proposition follow from the same fact: sum-product with a Gaussian kernel is the Gaussian chain's smoother with that kernel's covariance, a linear map, and the LMMSE of the true covariance is the best linear map in mean square; and the Gaussian log-likelihood of a noised sample is a function of its sample covariance, whose expectation $e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$ identifies (ρ, q).

A.2. Proposition 2

(i) $\mathbf{a}$ and $\mathbf{z}$ are independent and Gaussian, so $(\mathbf{a}, \mathbf{x})$ is jointly Gaussian with $\mathrm{Cov}(\mathbf{x}) = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I} = \mathbf{\Sigma}_t$ and $\mathrm{Cov}(\mathbf{a}, \mathbf{x}) = e^{-t}\mathbf{\Sigma}_0$. Hence $P_t = \mathcal{N}(0, \mathbf{\Sigma}_t)$, $S = \nabla \log P_t = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$, and $\langle \mathbf{a} \rangle_{x,t} = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\,\mathrm{Cov}(\mathbf{x})^{-1}\mathbf{x} = e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Then $\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t} = (\mathbf{\Sigma}_t - e^{-2t}\mathbf{\Sigma}_0)\mathbf{\Sigma}_t^{-1}\mathbf{x} = \Delta_t \mathbf{\Sigma}_t^{-1}\mathbf{x}$, which is Tweedie's identity $S = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$. (ii) Forward, by induction. $\overrightarrow{m}_1 = \mu = \mathcal{N}(0, 1)$ is the law of a_1 before any observation. If $\overrightarrow{m}_i(a) = p(a_i{=}a \mid x_{1:i-1}) = \mathcal{N}(a; m, P)$, Bayes' rule gives $\overrightarrow{m}_i \ell_i \propto p(a_i \mid x_{1:i})$, Gaussian with mean $m + k(x_i - e^{-t}m)$ and variance $(1 - ke^{-t})P$, $k = e^{-t}P/(e^{-2t}P + \Delta_t)$: the Kalman update. The Markov property then gives $\overrightarrow{m}_{i+1}(a') = \int K(a' \mid a)\overrightarrow{m}_i(a)\ell_i(a)\,\mathrm{d}a \propto p(a_{i+1}{=}a' \mid x_{1:i})$, Gaussian with mean ρ times the filtered mean and variance ρ^2 times the filtered variance plus q: the Kalman prediction. Backward, $\overleftarrow{m}_L \equiv 1$, and if $\overleftarrow{m}_{i+1}(a) \propto p(x_{i+2:L} \mid a_{i+1}{=}a)$ then $\overleftarrow{m}_i(a) = \int K(a' \mid a)\ell_{i+1}(a')\overleftarrow{m}_{i+1}(a')\,\mathrm{d}a' \propto p(x_{i+1:L} \mid a_i{=}a)$, again by the Markov property; it is the exponential of a quadratic in a (a Gaussian in information form, of zero precision at $i = L$). Since $x_{i+1:L}$ and $x_{1:i}$ are conditionally independent given a_i, $\overrightarrow{m}_i \ell_i \overleftarrow{m}_i \propto p(a_i \mid x_{1:i})\, p(x_{i+1:L} \mid a_i) \propto p(a_i \mid \mathbf{x})$. The Rauch–Tung–Striebel smoother computes the same conditional law by another factorisation, conditioning backward on a_{i+1}; two exact computations of one law agree, and its mean is $\langle a_i \rangle_{x,t}$ of (i). tests/test_kalman.py checks the three against each other to 10^{-12}.

A.3. Theorem 3

For $z \sim \mathcal{N}(0, 1)$, real a and $\alpha \in (0, 1)$, the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x)s^n/n!$ evaluated at $x = \alpha a + \sqrt{1 - \alpha^2}\, z$ and averaged over z gives $e^{s\alpha a - s^2\alpha^2/2} = \sum_n \mathrm{He}_n(a)(\alpha s)^n/n!$, hence $\mathbb{E}_z \mathrm{He}_n(\alpha a + \sqrt{1 - \alpha^2}\, z) = \alpha^n \mathrm{He}_n(a)$. The coordinates of $\mathbf{z}$ are independent given $\mathbf{a}$, so the product form holds for H_ν; integrating over $\mathbf{a} \sim P_0$ requires only $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$. The cases $n \leq 4$ are verified in Lean as polynomial identities under the first four Gaussian moments.

A.4. Theorems 5 and 6

Exact identities. With $\alpha = e^{-t}$ and $\Delta_t = 1 - \alpha^2$, Bayes' rule gives $P_t(\mathbf{x}) = \mathcal{N}_{\Delta_t}(\mathbf{x})\, \mathbb{E}_{P_0} \exp\bigl(\alpha \mathbf{x} \cdot \mathbf{a}/\Delta_t - \alpha^2|\mathbf{a}|^2/2\Delta_t\bigr)$, where $\mathcal{N}_{\Delta_t}$ is the $\mathcal{N}(0, \Delta_t \mathbf{I})$ density. Hence $\log P_t(\mathbf{x}) = \log \mathcal{N}_{\Delta_t}(\mathbf{x}) + \Lambda(s, b)$ with $s = \alpha\mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and $\Lambda(s, b) = \log \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$; $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] = \nabla_s \Lambda$ and $S = -\mathbf{x}/\Delta_t + (\alpha/\Delta_t)\nabla_s \Lambda$.

Where the expansion lives. Under Assumption 4 the integrand is dominated by $e^{\lambda|\mathbf{a}|}$ on $\{|s| \le \lambda\} \times \{b \ge 0\}$, so Λ is finite there and real-analytic in s on $|s| < \lambda$ for every $b \ge 0$, with Taylor coefficients at the origin equal to the joint cumulants of the linear form $s \cdot \mathbf{a}$ and the quadratic form $|\mathbf{a}|^2/2$; the tilted laws $\propto P_0(\mathbf{a}) e^{\theta(s\cdot\mathbf{a} - b|\mathbf{a}|^2/2)}$, $\theta \in [0,1]$, have moments of every order bounded uniformly on $\{|s| \le \lambda/2,\ 0 \le b \le 1\}$ by constants depending on λ and $\mathbb{E}e^{\lambda|\mathbf{a}|}$. Without Assumption 4 the expansion at $b = 0$ is not available: $\Lambda(s,0) = \log \mathbb{E}e^{s\cdot\mathbf{a}}$ is infinite for every $s \neq 0$ when P_0 has only finitely many moments.

Grading. Expand Λ in (s,b) at the origin and grade a monomial $s^j b^k$ by $j + 2k$; since $s = O(\alpha)$ and $b = O(\alpha^2)$, the grade is the order in α. By the symmetry of P_0 under $\mathbf{a} \mapsto -\mathbf{a}$ only even grades occur: $\Lambda = \Lambda_2 + \Lambda_4 + R_6$, with $\Lambda_2 = \frac{1}{2} s^\top \boldsymbol{\Sigma}_0 s - \frac{b}{2} \operatorname{tr} \boldsymbol{\Sigma}_0$ and

$$\Lambda_4 = \tfrac{1}{24}\kappa_4[s,s,s,s] - \tfrac{b}{4}\sum_i \operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i^2) + \tfrac{b^2}{8}\sum_{ij} \operatorname{cum}(a_i^2, a_j^2),$$

where cum is the joint cumulant under P_0 and the squares enter because b multiplies $|\mathbf{a}|^2/2$: $\operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i^2) = \operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i, a_i) + 2(\boldsymbol{\Sigma}_0 s)_i^2$ and $\operatorname{cum}(a_i^2, a_j^2) = \kappa_4[i,i,j,j] + 2(\boldsymbol{\Sigma}_0)_{ij}^2$, the added terms being functions of $\boldsymbol{\Sigma}_0$ alone. Every fourth-order joint cumulant of linear forms in $\mathbf{a}$ equals its value under the covariance twin plus a linear contraction of κ_4, so $\Lambda_4 - \Lambda_4^{\mathrm{G}}$ is linear in κ_4 and $\Lambda_2 = \Lambda_2^{\mathrm{G}}$. The remainder R_6 is the Lagrange remainder of grade six, a combination of sixth-order derivatives of Λ at an intermediate point of the segment to the origin, i.e. of tilted joint cumulants of order six; on the bulk $B = \{|s| \le \lambda/2\} = \{|\mathbf{x}| \le \lambda\Delta_t/2\alpha\}$ it is bounded by $C\alpha^6(1+|\mathbf{x}|)^6$ with C depending only on λ and $\mathbb{E}e^{\lambda|\mathbf{a}|}$, by the uniform moment bound above.

Leading term of the score gap. $S - S^{\mathrm{G}} = (\alpha/\Delta_t)\nabla_s(\Lambda - \Lambda^{\mathrm{G}}) = (\alpha/\Delta_t)\big[\nabla_s(\Lambda_4 - \Lambda_4^{\mathrm{G}}) + \nabla_s R_6 - \nabla_s R_6^{\mathrm{G}}\big]$. Substituting $s = \alpha\mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and expanding in α, the grade-four term is $\alpha^4 \nabla_{\mathbf{x}}\big[\frac{1}{24}\sum_{ijkl}\kappa_4[ijkl]\,{:}x_i x_j x_k x_l{:}\big] + O(\alpha^6(1+|\mathbf{x}|)^5)$, where $:\cdot:$ is the Wick (Hermite) product: the three contributions of Λ_4 combine into the Hermite form because $b = \alpha^2(1 + O(\alpha^2))$ and the cumulant contractions with δ_{ii} are exactly the terms that convert monomials into Hermite polynomials. The m-th component is $\alpha^4 \frac{1}{6}\sum_{jkl}\kappa_4[mjkl]\,{:}x_j x_k x_l{:}$.

Second moment. Under $\mathcal{N}$, Wick's theorem gives $\mathbb{E}_{\mathcal{N}}[{:}x_j x_k x_l{:}\,{:}x_{j'} x_{k'} x_{l'}{:}]$ equal to the number of perfect matchings of $\{j,k,l\}$ with $\{j',k',l'\}$ respecting indices; contracted with the symmetric tensor this yields $\mathbb{E}_{\mathcal{N}}\big[(\frac{1}{6}\sum_{jkl}\kappa_4[mjkl]{:}x_j x_k x_l{:})^2\big] = \frac{3!}{36}\sum_{jkl}\kappa_4[mjkl]^2$, and summing over m, $\mathbb{E}_{\mathcal{N}}\|\text{leading term}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2\alpha^8$. Replacing $\mathbb{E}_{\mathcal{N}}$ by $\mathbb{E}_{P_t}$ on the bulk costs a factor $1 + O(\alpha^2)$: $P_t/\mathcal{N} = 1 + \alpha^2\frac{1}{2}\langle \boldsymbol{\Sigma}_0 - \mathbf{I}, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle + O(\alpha^4(1+|\mathbf{x}|)^4)$ there, and the degree-2 correction carries an explicit factor α^2, so its contribution to the second moment is of relative size α^2. The cross term between the leading term and the grade-six remainder is $O(\alpha^{10})$ by Cauchy–Schwarz and the moment bound. The constant in $O(\alpha^2)$ depends on P_0 and L; for the Gaussian chain it is $\lambda_{\max}(\boldsymbol{\Sigma}_0)$ (last paragraph).

Tail. On $B^c = \{|\alpha\mathbf{x}| > \lambda\Delta_t/2\}$ the expansion is not used. By the tower property and Jensen, $\mathbb{E}_{P_t}[\|\mathbb{E}[\mathbf{a} \mid \mathbf{x}]\|^2 \mathbf{1}_{B^c}] \le \mathbb{E}[|\mathbf{a}|^2 \mathbf{1}_{B^c}]$, and $B^c \subset \{\alpha|\mathbf{a}| > c/\alpha\} \cup \{\sqrt{\Delta_t}|\mathbf{z}| > c/\alpha\}$ with $c = \lambda\Delta_t/4$, so $P_t(B^c) \le e^{-\lambda c/\alpha^2}\mathbb{E}e^{\lambda|\mathbf{a}|} + e^{-c^2/2\alpha^2}$, which is $\exp(-c' e^{2t})$ under Assumption 4. Since $\|S - S^{\mathrm{G}}\|^2 \le C(|\mathbf{x}|^2 + \alpha^2\|\mathbb{E}[\mathbf{a} \mid \mathbf{x}]\|^2 + \alpha^2\|\mathbb{E}^{\mathrm{G}}[\mathbf{a} \mid \mathbf{x}]\|^2)/\Delta_t^2$, Cauchy–Schwarz against $P_t(B^c)^{1/2}$ and the finiteness of $\mathbb{E}|\mathbf{a}|^4$ make the tail contribution below every power of α. Together, $\mathbb{E}_{P_t}\|S - S^{\mathrm{G}}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}(1 + O(e^{-2t}))$, which is Equation (7). Without Assumption 4 the tail probability is polynomial in α (of order $\alpha^{2\nu}$ for a tail index ν), and for $\nu \le 8$ the tail's own contribution is of order α^ν and leads; this is the heavy-tailed regime of Section 6.1.

Information and the general order. $\partial_\theta \log P_t = \partial_\theta \Lambda$. If $\partial_\theta \boldsymbol{\Sigma}_0 \neq 0$ the grade-two term $\frac{\alpha^2}{2}\langle \partial_\theta \boldsymbol{\Sigma}_0, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle$ leads, and $\mathbb{E}_{\mathcal{N}}$ of its square is $\frac{\alpha^4}{4} \cdot 2\|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$ by Wick, giving Equation (8). If $\partial_\theta \boldsymbol{\Sigma}_0 = 0$ the grade-four term $\frac{\alpha^4}{24}\sum \partial_\theta \kappa_4[ijkl]{:}x_i x_j x_k x_l{:}$ leads, with $\mathcal{N}$-second moment $\frac{4!}{24^2}\|\partial_\theta \kappa_4\|_F^2 \alpha^8$, giving Equation (9); the corrections and the tail are handled as above. For Theorem 6, the graded expansion of Λ and the expansion $\log\big(1 + \sum_{|\nu|\ge 2}\alpha^{|\nu|}\mathbb{E}_{P_0}[H_\nu]H_\nu(\mathbf{x})/\nu!\big)$ obtained from Mehler's kernel are two formal series in α for the same function whose coefficients at each grade are polynomials in the moments of P_0 up to that grade; they agree on every compactly supported P_0, where both converge, hence identically. The grade-m coefficient of $\partial_\theta \log(P_t/\mathcal{N})$ is therefore $\alpha^m \sum_{|\nu|=m} \partial_\theta \mathbb{E}[H_\nu]H_\nu(\mathbf{x})/\nu!$ plus ∂_θ of products of lower-grade terms, each of which vanishes by the

minimality of m; its $\mathcal{N}$-second moment is $\alpha^{2m}\sum_{|\nu|=m}(\partial_\theta \mathbb{E}H_\nu)^2/\nu!$ by orthogonality, and the bulk correction and the tail are as before. Since $:\mathbf{x}\mathbf{x}^\top:$ and the degree-four Hermite polynomials are $\mathcal{N}$-orthogonal, the leading cross-information between a degree-2 and a degree-4 coordinate vanishes, so the leading term of the efficient information of the degree-4 coordinate after degree-2 nuisances equals that of its raw information. For the Gaussian chain $\mathbf{x}_t \sim \mathcal{N}(0, \mathbf{\Sigma}_t(\theta))$ exactly and the stated trace formula is the Gaussian Fisher information; its limit $e^{4t} I_t \to \frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$ follows from $\mathbf{\Sigma}_t \to \mathbf{I}$ and $\partial_\theta \mathbf{\Sigma}_t = \alpha^2 \partial_\theta \mathbf{\Sigma}_0$, and is machine-checked in Lean (lean/README.md, target 3).

Corollary 7. At fixed $t > 0$ the family $\{P_t^\theta\}$ is differentiable in quadratic mean with finite, non-singular information (smoothness of Λ in θ under Assumption 4), so the local asymptotic minimax theorem of Hájek and Le Cam applies and the maximum-likelihood estimator is asymptotically efficient; the stated bound is $1/I_t^{\mathrm{eff}}$ with the efficient information of Theorem 6.

The crossover. For the Gaussian chain the expansion of $P_t/\mathcal{N}$ is an expansion in $\alpha^2\mathbf{\Sigma}_0$, i.e. in $\alpha^2\lambda_{\max}(\mathbf{\Sigma}_0)$ in operator norm, with $\lambda_{\max} \le (1+\rho)/(1-\rho)$ by the Toeplitz bound $\lambda_{\max} \le \sum_k \rho^{|k|}$; $\mathbf{\Sigma}_t^{-1}$ is within $\alpha^2\lambda_{\max}$ of $\mathbf{I}$ in operator norm, which is the source of $\epsilon_t = O(e^{-2t}\lambda_{\max})$ and of $t_* = \frac{1}{2}\log\lambda_{\max}$. The same chain has $P_t/\mathcal{N} \in L^2(\mathcal{N})$ iff $2\mathbf{I} - \mathbf{\Sigma}_t \succ 0$, i.e. iff $e^{2t} > \lambda_{\max} - 1$, so the Hermite expansion of $P_t/\mathcal{N}$ converges in $L^2(\mathcal{N})$ exactly above that time; for the Laplace chain it never does, which is why the proof goes through Λ.

B. Numerical representation and its control

The recursion. The two message sweeps are Equation (5) of Section 3; the site marginal is $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$ and Tweedie's identity turns its mean into the score.

Messages are represented on a uniform grid of N_{g} points on $[-A, A]$ with composite trapezoidal weights; the recursion is a scaled linear-domain forward–backward whose discarded constants are accumulated so the evidence $\log P_t(\mathbf{x})$ of the discretised model is exact to floating point. Three levels of exactness are kept apart: sum-product is exact at the level of functional messages; forward–backward is exact for the discretised factor model; and that model approximates the continuous chain with truncation (mass outside $[-A, A]$) and quadrature (second order in the spacing) errors that are measured, not assumed.

Validation where the answer is known. The Gaussian chain, where the closed form, the Kalman smoother and grid BP coincide (Proposition 2, Figure 4), is the sanity check of Section 4; it also reproduces brute-force enumeration of the discretised posterior on short chains to double precision (tests/test_gaussian_bp_equivalence.py, tests/test_fisher_bruteforce.py).

Against a finer reference ($N_{\mathrm{g}} = 1601$, $A = 12$) the worst relative score error over the twelve levels and four driving-noise families on ordinary draws is $9.9e-04$; under a three-standard-deviation displacement the heavy-tailed families lose two orders (1.7e-02). The worst forward-message boundary occupancy is $6.6e-05$ and the worst belief edge occupancy $2.9e-03$ over 60 production cells.

The envelope, measured rather than inherited. The validation above fixes the grid at the frozen configuration. Every other configuration this paper touches—the other driving-noise families, the correlations of the family ladder, the longer chains of the length study—inherits it, so we measured it instead: 5 driving-noise families × 4 correlations × twelve levels × 4 seeds, seven candidate grids against a reference at $N_{\mathrm{g}} = 3201$, $A = 12$.

At the frozen configuration the operating grid is comfortable: the pooled drift of the posterior-mean field is 9.2×10^{-6}, the worst single level 3.7×10^{-5}, and 2.8×10^{-4} of the belief mass lies in the outermost 2% of the domain. Nothing reported at that configuration is in question.

Over the whole sweep the picture is different, and the difference is instructive: at half-width 8 the worst-case drift is flat in the number of grid points, 5.4×10^{-2} at $N_{\mathrm{g}} = 201$ and 5.4×10^{-2} at $N_{\mathrm{g}} = 1601$, and falls only when the domain widens—2.6×10^{-2} at $(401, 10)$ and 3.0×10^{-3} at $(801, 12)$. That is the signature of truncation rather than of quadrature, and it means refinement in N_{g} alone, which is what refinement has meant in this literature, cannot fix it. The binding cell is the *student* family at $\rho = 0$, and the mechanism is a handful of sites: among the 8192 sites of each seed's bundle, $1, 1, 1, 2$ lie beyond $|a| = 8$ (the analytic expectation is 1.2), the

largest at $|a| = 15.4$. A site beyond the domain has its posterior pinned to the boundary on every grid, which is the "belief mass 1.00 in the outer 2%" of the diagnostic, and at the lowest level its score error alone is the drift of 5.4×10^{-2}. For a polynomial tail the expected exceedance count is Poisson with mean $ML\,P(|a| > A)$, which no finite domain makes zero: 0.17 per 8192 sites at $A = 12$, 0.04 at $A = 16$. Per replicate of the family ladder of Appendix E (512 chains of 32 sites) the same expectation is 2.39 sites at $A = 8$, 0.307 at 12 and 0.0655 at 16 for the Student-t_5 chain at $\rho = 0$, and below 0.188 for every other family and correlation; that ladder therefore quotes the heavy-tailed family from a recomputation on the $(1601, 16)$ domain, where its pooled floor has stopped moving (e07b, e07c); no claim in the body rests on it. The same consideration governs the closure panel of Figure 5: beyond $t \approx 3$ the reference on the $(801, 8)$ grid carries the tail of the truncated prior, and the heavy-tailed families depart from the constant for that reason alone; on $(3201, 32)$ they reach it (Appendix K).

The envelope also has to cover the longer chains of the length study and of the regime comparison, and it does (e11b: the same sweep at $L \in \{64, 128\}$, 4 seeds). On the operating grid the worst pooled drift over the sweep is 5.4×10^{-2} at $L = 32$, 5.8×10^{-2} at 64 and 6.6×10^{-2} at 128, a factor of at most 1.22, and at the frozen configuration 9.2×10^{-6}, 7.2×10^{-6} and 4.5×10^{-6}: the per-site quadrature error does not compound along the chain, because the messages are renormalised at every site.

A second observation from the same sweep cuts the other way. The column-mass defect of the discretised kernel, $\max_j |\sum_k K(u_k \mid u_j) w_k - 1|$, grows with correlation, from 7.0×10^{-3} at $\rho = 0$ to 0.15 at $\rho = 0.95$, and the reason is again the domain and not the spacing: the maximum sits in the outermost columns, where a kernel centred at ρu_j with $|u_j|$ near A loses the part of its mass beyond the domain, so the defect is flat in N_{g}, falls as the domain widens, and is at machine precision in the interior columns at every ρ. It is not predictive of the error: at $\rho = 0.95$ the drift is 1.2×10^{-2}, because the recursion renormalises every message and the affected columns carry almost no belief. We report it as a diagnostic and not as an error bar. The uniform family is the one exception to the quadrature picture: its kernel is discontinuous, the trapezoidal rule is first-order on it, and its worst-level drift is largest on the coarsest spacing rather than the narrowest domain.

Refinement on the reported estimands. Re-scoring the validation-selected kernels of the capacity experiment on their own test bundles at $N_{\mathrm{g}} = 801$ and 1601 ($A = 8$) and at $A = 10$ ($N_{\mathrm{g}} = 801$), with the reference recomputed at each resolution, moves the pooled error of EM–BP ($C = 8$) by at most 0.00%, 0.00% and 0.00% at $M = 2048$ (maximum over 16 seeds), and the LMMSE floor by at most 0.00%; the reference score itself moves by 0.001% between $N_{\mathrm{g}} = 401$ and 801. The reported effects are three orders of magnitude larger than the discretisation.

Floors in the theory verification. The outer-product information estimate of a shape coordinate is a small residual of order-one quantities; for the kinked Laplace density its quadrature floor is reached near $t = 2.5$ at $N_{\mathrm{g}} = 801$ (the replicate spread stops shrinking), which is why Figure 5b reports the asymptotic slope on the smooth members of the family. The closure gap of the uniform family reaches the double-precision floor of the BP means near $t = 3.5$, where its squared gap is 10^{-13} per chain.

C. Experimental protocol and provenance

Frozen configuration. Laplace chain unless stated, $\rho = 0.85$, $q = 1 - \rho^2$, $a_1 \sim \mathcal{N}(0, 1)$, $L = 32$; grid $N_{\mathrm{g}} = 401$, $A = 8$; twelve log-spaced levels $t \in [0.05, 3.0]$; other families Gaussian, Student-t ($\nu = 5$), uniform, symmetric two-component mixture ($\kappa = 0.6$), generalised Gaussian ($\beta \in [1, 6]$), all variance-matched. Sixteen confirmatory seeds 0–15 for every comparison at the frozen configuration; eight for the regime sweeps; development seeds 10000–10003 for every architecture and hyperparameter choice (three restarts of 32,000 steps per configuration at $M \in \{32, 2048\}$, ranked by pooled bulk risk; Table 6). Validation and test bundles are drawn per seed (1024 and 2048 sequences) from streams disjoint from every training stream; their hashes are stored with every row so pairing across arms is checkable.

Estimand. Schedule-pooled relative score error $\left(\sum_t \sum_{j,i} (\hat{s}_t(\mathbf{x}_j)_i - s_t(\mathbf{x}_j)_i)^2 / \sum_t \sum_{j,i} s_t(\mathbf{x}_j)_i^2\right)^{1/2}$ on all sites (the headline), on the interior sites 5–28 (used for selection), and on the centre site (diagnostic). Ratios are formed within seed and averaged; intervals are percentile bootstraps over seeds.

Table 6. Architecture screen under one recipe on the development seeds (never used for a reported number): the best configuration per family, its parameter count, its pooled bulk risk over 4 development seeds, two sizes and 3 restarts of 32,000 steps per configuration, and the risk of the family's runner-up configuration.

architecture	selected configuration	parameters	screen risk	runner-up
MLP (global)	hidden: (128, 128), lr 0.001, eps	26,016	0.390	0.411
window head	radius: 4, width: 64, lr 0.001, eps	5,505	0.116	0.122
dilated TCN	channels: 32, dilations: (1, 2, 4, 8), lr 0.001, eps	17,057	0.143	0.160
bidirectional GRU	hidden: 32, lr 0.001, eps	15,169	0.127	0.133
transformer	d: 32, heads: 4, layers: 2, lr 0.003, eps	18,057	0.147	0.154
1-D U-Net	channels: 16, lr 0.001, eps	13,713	0.148	0.212

Arms and interfaces. EM–BP: mixture kernel, C as stated, ρ initialised at 0.3, one noised view per sequence with sequences split evenly across the twelve levels (interface C), checkpoint ladder $\{10, 20, 40, 60, 80, 120, 160, 220, 300, 400, 600, 800, 1200\}$ selected on the validation bundle scored against the true kernel; selecting instead on held-out one-view log-evidence, which needs no true score, gives the same test error within seed noise (Appendix K). Networks under interface A: trained by denoising score matching on the clean sequences with the level and the corruption redrawn at every step, batch 64, Adam, gradient clipping at 1, both output parameterisations, checkpoint ladder to 32,000 steps selected on the validation bundle. Interface B: the same objective on one fixed corruption per clean sequence, the checkpoint selected as under interface A on the validation bundle scored against the true kernel. Interface C: implicit score matching on the same one-view noisy bundle as EM–BP, with the divergence computed exactly for the window head and by Hutchinson's estimator otherwise, and the checkpoint selected on the implicit objective of a held-out noisy bundle; the true score enters only in the final scoring. The misspecification experiment's network is a radius-2, width-64 window head trained for 16,000 steps under interface A.

Provenance. Every run directory carries the commit, the dirty-tree list on the code path, the resolved configuration digest, the seeds, host and environment; a generator refuses to emit a table from a run whose code path was dirty or from an incomplete design, and flags, without dropping, any cell whose narrowest fitted component is under two grid cells. Every number in this paper is a macro emitted by such a generator; the manuscript check (tools/check_manuscript.py) fails the build on any typed decimal, and an independent re-derivation of every headline number from the frozen files (tools/audit_claims.py) runs with it. The regime sweep was recomputed at twice the seed count in a fresh output directory, from a different deployment of the source and on a different compute node; across the 480 rows it shares with E13, every risk and every fitted moment agrees to the last bit (maximum absolute difference 0 in the pooled error and 0 in the fitted excess kurtosis).

D. The oracle ladder

Table 7. Schedule-pooled relative score error of the two oracles against the tree-exact reference, 512 chains per family at the frozen configuration. The LMMSE column is the floor of every second-order estimator (Proposition 1).

family	excess kurtosis	isotropic $-\mathbf{x}$	LMMSE, true $\mathbf{\Sigma}_0$
Laplace	+3	0.670	0.168
Student-t_5	+6	0.668	0.158
uniform	−1.2	0.671	0.149
bimodal	−0.72	0.663	0.082
Gaussian	0	0.662	0.000

E. The second-order floor across correlation

The frozen configuration fixes $\rho = 0.85$. The price of second-order closure is a function of ρ as well as of the driving-noise law, and this appendix reports it over $\rho \in \{0, 0.3, 0.6, 0.85, 0.95\}$ for five families on the twelve-level schedule, computed from the exact score with no estimation involved. The heavy-tailed family is quoted from a

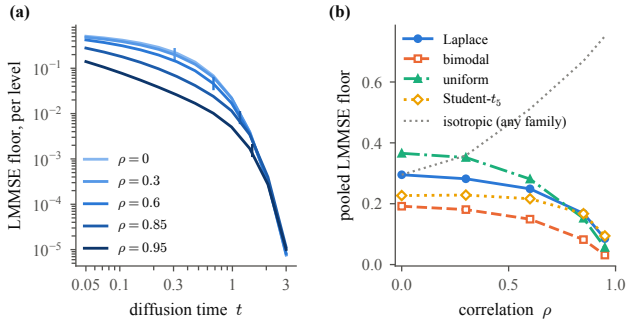


Figure 16. The second-order floor across correlation. (a) Per-level LMMSE floor on the Laplace chain for five ρ (ticks: $t_*(\rho)$); the floor collapses after the crossover. (b) Schedule-pooled floor against ρ for four driving-noise families, with the isotropic reference: strong correlation makes the score energy second-order-dominated, so the relative price of closure falls with ρ while the absolute fourth-order structure does not. The Student-t_5 curve is computed on the $(1601, 16)$ domain, where its $\rho = 0$ cell is converged (Appendices B and E); the other families on the operating grid.

recomputation on the $(1601, 16)$ domain (e07c): on the operating grid the quadrature sweep of Appendix B places its $\rho = 0$ cell outside the validated envelope, one to two sites per bundle lying beyond the domain and pinned to its boundary. Widening the domain moves its pooled floor by at most 0.005 from $(801, 8)$ to $(801, 12)$ and by at most 0.0002 from $(801, 12)$ to $(1601, 16)$ (e07b, e07c), and every other family by at most 0.0010 between the operating and the widest grid; the other families are quoted from the operating grid, where they are converged.

Across correlation. The floor is a function of the driving-noise law and of ρ (Figure 16): on the Laplace chain it is 0.295 for independent sites, 0.168 at the frozen $\rho = 0.85$ and 0.085 at $\rho = 0.95$, where $t_* = 1.50$ and the covariance carries most of the score energy until late in the schedule. The families order by the size of their fourth cumulant at every ρ; the isotropic reference rises with ρ because the covariance itself is what an unstructured estimator must learn first.

F. Capacity: the previous reading, and why it is void

An earlier study of this model compared mixture kernels of $C \in \{1, 2, 4, 8, 16\}$ at $M \in \{128, 512\}$ over sixteen seeds and concluded that no capacity improved on a single linear-Gaussian kernel, attributing the estimator's margin over the networks to the Markov factorisation and its covariance alone. That sweep is void for a reason prior to any reading of it: its training draw re-created the random generator for every sequence, so every arm at every capacity and seed was fitted to a single chain of L sites (its validation and test bundles were drawn correctly). The signature is in its own numbers: Proposition 1 places a correctly fitted $C = 1$ kernel on the LMMSE floor, 0.168 on the Laplace chain, where Section 6 and all fifteen cells of the regime sweep land to 0.0099, and the earlier sweep's single-Gaussian arm scatters across a range several times wider. The stopping asymmetry that sweep also shows (its likelihood tolerance fired for almost every single-Gaussian cell and for no cell with $C \geq 4$, so that the richer arms reported checkpoints far past their optimum) is the behaviour of em on one chain, not evidence about capacity in either direction; we had first read it as the explanation of the earlier conclusion, and withdraw that reading. The capacity experiment of Section 6 draws every chain independently, asserts it, and selects the checkpoint identically for every capacity on a per-seed validation bundle; the same defect was found and fixed in one of our own drivers before any of its numbers reached a page (Appendix C).

G. Capacity across the driving-noise family

Moving the driving noise from Laplace (excess kurtosis +3) through the Gaussian to the platykurtic end at fixed ρ and variance (Figure 17), the LMMSE floor traces a V: 0.166 at the Laplace end, exactly zero at $\beta = 2$ where the covariance twin is the truth, and 0.119 at the platykurtic end. The linear-Gaussian kernel follows that V within its own standard error at every member, which is Proposition 1 read across a family rather than at a point; the mixture kernel stays between 0.017 and 0.035 throughout. The paired gain of $C = 8$ over $C = 1$ is therefore ordered by κ_4: 0.131 ($[0.127, 0.135]$) at Laplace, 0.075 at $\kappa_4 = +1.53$, and 0.090 at the platykurtic end. At the

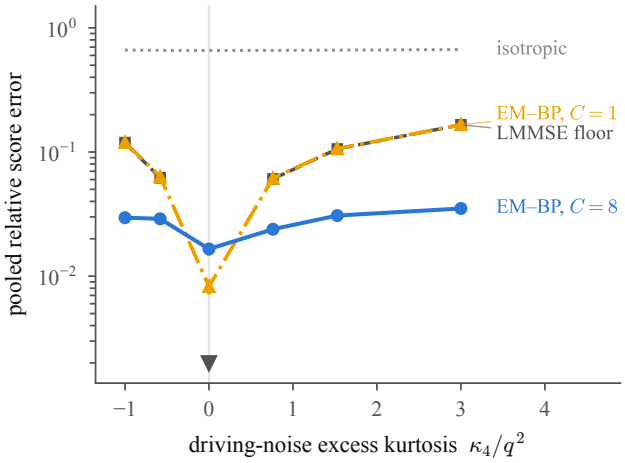


Figure 17. Capacity buys exactly what the closure discards. Pooled relative score error against the driving noise's excess kurtosis for the generalised-Gaussian family at fixed ρ and variance ($M = 512$, 16 seeds, $\pm$ s.e.): the linear-Gaussian kernel tracks the LMMSE floor, which is the exact price of second-order closure for each member; the mixture kernel tracks the exact score, and the two meet at the Gaussian member, where there is nothing beyond second order to learn. At that member the floor is exactly zero—the covariance twin is the truth—which a log axis cannot show; the line is broken there and the caret marks the point.

Gaussian member the gain reverses: -0.008 ($[-0.011, -0.005]$), a small but unambiguous loss, because there is no fourth-order structure to buy and the extra components only add estimation variance. Non-Gaussian capacity is worth what Theorem 5 says it is worth—the fourth-order structure the covariance twin leaves behind—and nothing where the twin leaves nothing.

H. The capacity contrast across regimes

The capacity result above is measured at one point of the model's parameter space, so we re-ran the contrast that carries it across 15 regimes: correlations from 0.3 to 0.95 and chain lengths from 16 to 64, everything else at the frozen protocol, 16 seeds each. The mixture beats the Gaussian kernel in *all* cells with the paired interval excluding zero, by -0.253 at the most favourable corner and -0.047 at the least. The sweep also checks the apparatus at every point rather than only at the frozen configuration: Proposition 1 says the $C = 1$ arm must coincide with the LMMSE of the true covariance, and it does, to 0.0099 in the worst of the 15 cells, with a fitted excess kurtosis of at most 0.000 in absolute value. The $C = 8$ arm recovers the driving noise's true excess kurtosis of 3 everywhere (2.92 to 3.31 across correlations), and its spread widens from ± 0.33 to ± 1.27 as ρ rises—the shape becomes harder to see, not differently placed.

The advantage itself shrinks monotonically with ρ, which is the direction Section 6.1 gives: a strongly correlated chain has a nearly Gaussian marginal, the stationary marginal's excess kurtosis falling as $3(1-\rho^2)/(1+\rho^2)$ from 2.50 at $\rho = 0.3$ to 0.15 at $\rho = 0.95$, so there is less fourth-order structure for capacity to buy. We report this as a direction and not as a rate: the pooled error runs over a schedule whose informative window moves with ρ, so its trend is not the exponent of Theorem 5 and we do not fit one to it.

I. Generation and length extrapolation: the tables

J. The matched comparison across regimes

The arms and selection rules are those of Section 6 with two additions: the evidence-selected EM–BP row, which needs no true score (Appendix K), and the network-favourable interface B row. A network cell is excluded from the factor only if its selected held-out implicit objective is not below that of the trivial score $-\mathbf{x}$ on the same views, which is what a fit that diverged looks like on its own criterion; no cell is excluded (0 network cells fail the test), and 6 carry the curve-shape flag. The two EM–BP rows agree: the same rung is selected in 52 of 96 cells and the largest difference in test error is 0.019; the last rung of the ladder is selected in 68 of 96 oracle-

Table 8. Generated samples against the true law at $t_{\min}$: mean log-likelihood gap $\Delta\ell$ to the reference sample under the true $P_{t_{\min}}$ (exact evidence; the exact arm's gap is the sampler's discretisation floor), driving-noise excess kurtosis $\hat{\kappa}_4$, lag fourth cumulant, sliced W_2; the row marked reference carries the reference sample's own statistics. Mean over seeds.

M	arm	$\Delta\ell$	$\hat{\kappa}_4$	lag-κ_4	W_2
128	reference	–	1.94	0.000	–
	exact	0.074	1.86	0.000	0.051
	em–bp	0.204	1.68	0.000	0.078
	lmmse	1.761	0.00	0.000	0.053
	isotropic	46.241	0.00	0.006	0.244
	net:bigru	0.752	1.10	0.006	0.160
	net:window	1.284	1.27	0.004	0.154
2048	reference	–	1.94	0.000	–
	exact	0.070	1.87	0.000	0.051
	em–bp	0.164	1.80	0.001	0.055
	lmmse	1.766	0.00	0.000	0.053
	isotropic	46.295	0.00	0.010	0.244
	net:bigru	0.385	1.55	0.003	0.167
	net:window	1.115	1.46	0.003	0.139

Table 9. Length extrapolation at $M = 2048$: pooled relative score error of estimators fitted at $L = 32$ and evaluated against the tree-exact score of longer chains, mean over 16 seeds. Longer chains are slightly easier (proportionally fewer boundary sites), so a length-agnostic arm should improve a little; the transformer instead breaks down beyond its training length.

arm	$L = 32$	$L = 64$	$L = 128$	$L = 256$
bi-GRU	0.060	0.057	0.057	0.056
EM–BP	0.023	0.023	0.023	0.023
TCN	0.068	0.066	0.065	0.064
transf.	0.063	0.243	0.514	0.703
U-Net	0.066	0.064	0.062	0.062
window	0.118	0.099	0.088	0.082

selected cells and 56 of 96 evidence-selected ones, so the EM–BP error is an upper bound there and every factor is conservative. Against the second-order floor, tested per seed on the same test bundle, the best network sits below it in $(0.5, 32), (0.5, 128), (0.85, 128)$ and above it in $(0.85, 32), (0.95, 32), (0.95, 128)$. At the frozen point it is above by $+0.003$ [$+0.000$, $+0.005$], a ratio of 1.02: at matched information the best network reaches the floor there and does not pass it, where with clean sequences and a fresh corruption at every step 5 of six architectures pass it (Section 6). At $\rho = 0.95$ the margin is $+0.110$ against exact radius-4 and radius-8 floors of 0.072 and 0.025 (Appendix K): the miss is the architecture's, not the window's. At the frozen point the factor is 8.3 with per-seed factors from 4.1 to 19.9, against 7.9 with per-seed factors from 5.1 to 14.6 over the sixteen seeds of Section 6: the same population at a different seed count. The networks' error falls with L because each noised view carries proportionally more sites; so does EM–BP's, and the factor survives.

Platykurtic driving noise (e14b). Every matched comparison above uses the Laplace driving noise, whose excess kurtosis is positive, so the advantage could in principle be a statement about heavy tails. The frozen cell repeated with the symmetric two-component mixture, whose excess kurtosis is negative at the same covariance, answers it: the second-order floor is 0.082, well below the Laplace value, because there is less fourth-order structure for the closure to discard; the best network, here the transformer, reaches 0.159, close to its Laplace error, and therefore sits a factor 1.93 above the floor rather than at it; EM–BP reaches 0.021, a paired factor 8.8 [7.4, 10.2] in 16 of 16 seeds, a factor indistinguishable from the Laplace factor at the same point. The estimator's margin is the fourth-order structure the covariance twin leaves behind, of either sign.

K. Reference computations behind the body

Six deterministic or cheap computations support statements in the body; each is a registered experiment (e15–e21) whose outputs are frozen like every other.

Table 10. Interface C across correlation and length (e14): pooled relative score error of the LMMSE oracle, of EM–BP with its iteration count chosen by held-out one-view log-evidence (ev.) and by validation risk against the true kernel (bp), of the best network trained by implicit score matching on the same one-view bundle (C) and of the best network given the clean sequence with one fixed corruption (B), each re-screened per regime on two development seeds; last column the paired factor of the C network over the evidence-selected EM–BP with a percentile bootstrap over 16 seeds. $^{\dagger n}$: n cells whose held-out objective never improved on its first checkpoint (reported, not excluded).

ρ	L	lmmse	EM–BP (ev.)	EM–BP (bp)	best net (C)	best net (B)	factor C / EM–BP (ev.)
0.5	32	0.265	0.026	0.024	0.137	0.129	5.7 [4.9, 6.7]
0.5	128	0.268	0.020	0.018	0.089	0.079	4.6 [4.1, 5.0]
0.85	32	0.167	0.024	0.021	0.170	0.156	8.3 [6.6, 10.3]
0.85	128	0.168	0.015	0.015	0.106	0.095	8.0 [6.7, 9.2]
0.95	32	0.086	0.021	0.019	0.196	0.128	10.3 [8.8, 11.9]$^{\dagger 6}$
0.95	128	0.087	0.017	0.016	0.105	0.094	6.4 [6.0, 6.9]

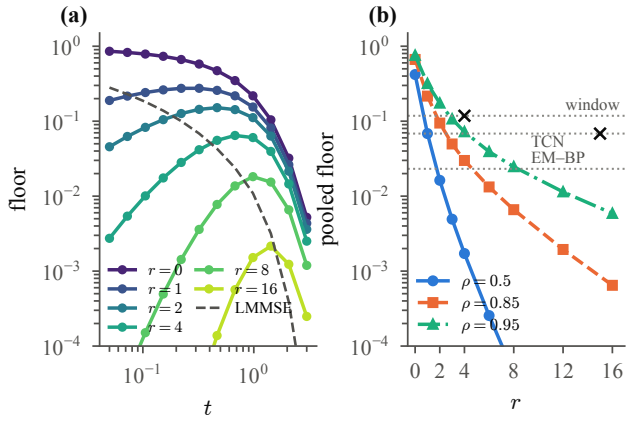


Figure 18. Locality against its floor. (a) Relative score error of the exact radius-r posterior mean on the Laplace chain at $\rho = 0.85$ at every level of the schedule, with the Gaussian twin's full LMMSE for scale: at low noise a window of one site is nearly enough, at $t \approx 1$ a radius of 16 still costs a thousandth. (b) The floor pooled over the schedule against the radius for the three sweeps; dotted lines mark the frozen all-site errors at $M = 2048$ of the radius-4 window head (the screened winner), the dilated TCN (receptive radius 15) and EM–BP, the crosses their radii. The window head sits four times above its floor; the TCN sits well above the floor of any radius past 8.

The locality floor (e15). For the Laplace chain at $\rho \in \{0.5, 0.85, 0.95\}$ and every level of the schedule, the best estimator of a_i from the window $x_{i-r}, \dots, x_{i+r}$ is its posterior mean given the window, computed exactly by sum-product on the sub-chain with the exact marginal law of the window's first site (propagated on the grid) as initial factor; its relative score error against the full-chain score is a floor for every radius-r denoiser. At $\rho = 0.85$ the pooled floors are 0.666, 0.094, 0.030, 0.007 and 0.001 at $r = 0, 2, 4, 8, 16$; the Gaussian twin's windowed LMMSE is 0.173 at $r = 4$ against its full value 0.170, so the non-Gaussian part of the score is what a wider window carries. At $\rho = 0.5$ the radius-4 floor is 0.002. For the Gaussian chain the score's dependence on a site at distance k decays as $r(\rho, t)^k$ with $r = (A - \sqrt{A^2 - B^2})/B$, $A = \Delta_t(1 + \rho^2) + e^{-2t}(1 - \rho^2)$, $B = 2\Delta_t\rho$, from 0 at $t = 0$ to ρ as $t \to \infty$.

The fourth-order correction (e16). With $y = \mathbf{\Sigma}_t^{-1}\mathbf{x}$, $K = e^{-4t}\kappa_4$ and $A_{ml} = \sum_{jk} K_{mjkl}(\mathbf{\Sigma}_t^{-1})_{jk}$, the first Edgeworth correction around the twin is $\epsilon(\mathbf{x}) = \frac{1}{24}\big[K[y, y, y, y] - 6\, y^\top A y + 3\langle A, \mathbf{\Sigma}_t^{-1}\rangle\big]$ and the corrected score $S^{\mathrm{G}} + \nabla\epsilon/(1+\epsilon)$ with $\nabla\epsilon = \frac{1}{6}\mathbf{\Sigma}_t^{-1}K[\cdot, y, y, y] - \frac{1}{2}\mathbf{\Sigma}_t^{-1}Ay$. Fraction of the squared closure gap closed on the Laplace chain: 0.300 at $t = 0.22$, 0.826 at $t = 0.32$, 0.961 at $t = 0.68$, 0.985 at $t = 0.98$, 1.000 at $t = 2.07$; the bimodal and uniform chains cross ninety per cent at $t = 0.47$ and 0.47. Pooled over the schedule the corrected score is worse than the twin, because the lowest levels dominate the pooled energy and the expansion diverges there: the correction is a statement about what the gap is made of at each level, not a competitor on the schedule.

Closed-form misspecification bias (e17). For a Gaussian truth with covariance $\mathbf{\Sigma}(\delta)$ the population limit of the linear-Gaussian fit under the one-view protocol is $\theta^*(\delta) = \arg\min_{(\rho,q)} \sum_t \mathrm{KL}(\mathcal{N}(0, \mathbf{\Sigma}_t(\delta)) \,\|\, \mathcal{N}\Big(0, \hat{\mathbf{\Sigma}}_t(\rho, q)\Big))$ and its pooled relative score error is $b(\delta)^2 = \sum_t \mathrm{tr}[(\hat{\mathbf{\Sigma}}_t^{-1} - \mathbf{\Sigma}_t^{-1})\mathbf{\Sigma}_t(\hat{\mathbf{\Sigma}}_t^{-1} - \mathbf{\Sigma}_t^{-1})] / \sum_t \mathrm{tr}\, \mathbf{\Sigma}_t^{-1}$. Against the frozen EM–BP errors at $M = 2048$: the second-lag coefficient at the largest strength 0.247 (KL projection), 0.216 (lag-one plug-in) versus 0.214 measured; site-varying ρ at its largest strength 0.268 versus 0.287; long-range coupling 0.235, 0.202 versus 0.204; rank-one latent at $\delta = 1$ 0.065 versus 0.052. The largest deviation of the lag-one projection from the measured error over all cells is 0.019. Small-δ laws: $b \approx 0.60\,\delta$ (second lag), $1.75\,\delta$ (site-varying), $1.07\,\delta$ (long-range, local exponent 0.79), $0.15\,\delta^2$ (rank-one).

The closure constant and the heavy-tailed regime (e18). The per-sequence squared gap is a degree-six polynomial in nearly Gaussian variables and heavy-tailed (maximum over 8192 sequences 283 times the median): the relative standard error of the mean is 4% from 8192 sequences, hence about four times that from 512. The constant is therefore quoted from 8192 sequences with a bootstrap interval: 0.92 $[0.84, 1.01]$ at $t = 3$ and 0.99 $[0.91, 1.08]$ at $t = 4$ for Laplace. Across grids, with paired sequences, Student-t_5's local slope on the half-unit interval from $t = 4$ is -4.03 at $(801, 8)$, -5.92 at $(1601, 16)$ and -7.92 at $(3201, 32)$, against -8: the departure on the narrow domain is the truncated prior in the reference. Student-t_7 and t_9 behave the same way. The uniform family departs identically on every grid from $t \approx 3.5$: that is the double-precision floor of the posterior means, reached by every family near $t = 5$.

Moment estimators (e19) and selection without the true score (e20). On e06's data the moment estimator of ρ has RMSE 0.031, 0.039, 0.060 and 0.385 at $t = 0.05$, 0.4, 1 and 2; that of κ_4 has RMSE 1.42, 1.84 and 15.22 at $t = 0.05$, 0.4 and 1, with a bias of -0.75 at $t = 0.4$. Selecting EM–BP's checkpoint by held-out one-view log-evidence instead of by validation risk against the true kernel gives a test error of 0.105 against 0.105 at $M = 32$, the size at which selection matters most (7 of 8 seeds select the same rung; at $M = 2048$ the validation curve is monotone through the ladder, so every rule picks its top).

The information curve at $\rho = 0.5$ and the scalar constants (e21). At $\rho = 0.5$ the local slope of $\log(1/I^{\mathrm{eff}})$ on the levels $[0.6, 1.0]$ is 6.1–6.9 for the shape coordinate and 3.8–3.8 for the correlation, against the asymptotic 8 and 4: a sample-complexity experiment on those levels must be read against the curve, not the asymptote. The Cramér–Rao reference for the excess kurtosis at $t = 1$ is 1.62 at $M = 2048$ and 0.81 at $M = 8192$. In the scalar generalised-Gaussian family an untruncated quadrature gives $I_t(\beta)e^{8t}$ over $\frac{1}{24}(\partial_\beta \kappa_4)^2$ equal to 1.0000 ($\beta = 2$), 1.0009 ($\beta = 3$) and 0.9994 ($\beta = 1.5$) at $t = 2$, and the variance coordinate's $I_t e^{4t}$ equal to 1.0000 times one half.

Information in cumulant coordinates. The information curves of the generalised-Gaussian chain are measured in the coordinates (ρ, β) in which the chain is parameterised. Fisher information transforms as a tensor, so the information about the driving noise's second and fourth cumulants, $\kappa_2 = q$ and $\kappa_4 = \gamma_2(\beta)q^2$, is obtained by the delta method from the efficient information about (ρ, β) and the Jacobian of $(\rho, \beta) \mapsto (\kappa_2, \kappa_4)$; it is not a relabelling of the β curve. Near the Gaussian the two differ by orders of magnitude: from $\beta = 1$ to $\beta = 3$ the information about β falls by a factor of several hundred while the information about κ_4 rises, because β barely moves any cumulant there, and a β curve labelled as fourth-order information would be wrong by a factor between 1.6 and 1526 over the schedule. In these coordinates the ratio of the fourth- to the second-order information at the Laplace member is 0.071 at $t = 0.05$ and 9.9×10^{-5} at $t = 3$, a decay of fourth-order relative to second-order information by a factor 717 over that range (the ratio at $t = 4$, 4.3×10^{-4}, is at the numerical floor of the shape estimator). The fitted slopes in Figure 5b are for $t \geq t_* + 1$ on the mean over replicates; for the Laplace and $\beta = 1.25$ members the fourth-order slope (-2.54, -2.53) is a floor artefact, as the β slope was (Appendix B), and the rate is read on the smooth members ($\beta = 2, 3$).

L. The ϕ^4 field in detail

The chain. With the site terms split symmetrically, the lattice energy of Section 2 defines the symmetric, positive, Hilbert–Schmidt transfer operator $\mathsf{T}(\phi, \phi') = e^{-hV(\phi)/2\Theta} e^{-(\phi'-\phi)^2/2\Theta h} e^{-hV(\phi')/2\Theta}$. Its largest eigenvalue λ_0 is simple with a positive eigenfunction ψ_0 (Perron–Frobenius), and the infinite-volume Gibbs measure is the stationary chain with marginal $\pi = \psi_0^2$ and kernel $K(\phi' \mid \phi) = \mathsf{T}(\phi, \phi')\psi_0(\phi')/(\lambda_0 \psi_0(\phi))$, reversible with respect

Table 11. The site energy (five parameters) fitted through noise, read as physics: temperature $\hat{\Theta}/\Theta$ and barrier height of the fitted kernel at $M = 2048$ (mean over sixteen seeds, standard deviation in brackets; exact values 1 and 0.25), the number of fits whose barrier is within 20% of the exact one (band, read where $\Theta \le 1$), and the risk divided by the linear floor at each sample size.

	$M = 2048$			risk / floor			
cell	$\hat{\Theta}/\Theta$	barrier	band	32	128	512	2048
$\Theta = 0.3$, $h = 1$	1.000 (0.023)	0.247 (0.011)	16/16	0.272	0.139	0.067	0.033
$\Theta = 0.5$	1.005 (0.012)	0.246 (0.015)	16/16	0.657	0.269	0.159	0.072
$\Theta = 1$	1.005 (0.021)	0.248 (0.036)	14/16	1.367	0.534	0.278	0.141
$\Theta = 2$	1.000 (0.013)	0.277 (0.083)	–	1.429	0.630	0.414	0.169
$\Theta = 0.3$, $h = 0.5$	1.007 (0.008)	0.254 (0.007)	16/16	0.277	0.136	0.063	0.040
$\Theta = 0.3$, $h = 0.25$	1.001 (0.003)	0.250 (0.007)	16/16	0.237	0.130	0.052	0.032

to π. The lag-k covariance is $\sum_n (\lambda_n/\lambda_0)^k \langle\psi_0|\phi|\psi_n\rangle^2$, so λ_1/λ_0 plays for the field the role ρ plays for the linear chain, and $\xi = 1/\log(\lambda_0/\lambda_1)$. We diagonalise T densely on 2401 points on $[-12, 12]$; samples are exact in the continuum (a rejection sampler on the tilted walk) and on the lattice (an inverse-CDF sampler), and the two agree.

The tilted-walk identity and reversibility. Rearranging the eigen-equation, $K(\phi' \mid \phi) = \mathcal{N}(\phi'; \phi, \Theta h)\, g(\phi')/Z(\phi)$ with $g = \psi_0 e^{-hV/2\Theta}$ and $Z(\phi) = \lambda_0 \psi_0(\phi) e^{hV(\phi)/2\Theta}$: a Gaussian step tilted by a function of the arrival point, with the column normaliser the h-transform. Every tilted walk $K_g(a' \mid a) = \mathcal{N}(a'; a, \gamma)\, g(a')/Z_g(a)$ is reversible with invariant density $\pi_g \propto g Z_g$, since $\pi_g(a) K_g(a' \mid a) \propto g(a) g(a') \mathcal{N}(a'; a, \gamma)$ is symmetric. It is the Gibbs chain of the symmetric pair energy $W(a, a') = (a' - a)^2/2\gamma + f(a) + f(a')$ with $f = -\frac{1}{2}\log(g/Z_g)$, and conversely; so the directed kernel and the symmetric pair energy are one model class. For the field's own (γ, g), $f = (h/2\Theta)V(sa)$ up to a constant, $\hat{\Theta} = \hat{\gamma} s^2/h$ reads the temperature, and $\hat{V}(\phi) = (2\hat{\Theta}/h)\hat{f}(\phi/s) + \text{const}$ reads the potential, where $s^2 = \mathrm{Var}_\pi\, \phi$.

Where the linear chain and the field meet, and why the mixture kernel fails. A Gaussian linear chain with $0 < \rho < 1$ is the member of the tilted family with $\log g$ quadratic (step variance q/ρ, tilt variance $1 + \rho$); no other linear chain is, and no other tilted walk has state-independent driving noise, so the two families intersect at the Gaussian chain only. The mixture kernel is misspecified on the field through its drift: the conditional mean step of a tilted walk is exactly $\gamma(\log Z)'(a)$, the gradient of the log-normaliser, which for the field is $\Theta h(\log \psi_0)'(\phi) + (h^2/2)V'(\phi)$, a nonlinear function with its zeros inside the wells, while the mixture's drift is linear. The misspecification is therefore through the drift, not through the shape of the driving noise, which is a different mechanism from every Gaussian departure of Figure 15.

Physical statistics. A kink is a sign change that survives merging every run shorter than one kink width $\sqrt{2}/h$ sites; a domain is the run between two kinks; the bimodality coefficient is $(\text{skew}^2 + 1)/(\text{excess kurtosis} + 3)$. The raw sign-change density per unit length grows like $h^{-1/2}$ under refinement (0.120, 0.176, 0.247 at $h = 1, \frac{1}{2}, \frac{1}{4}$) and is not a physical quantity; the coarse density is constant across the three refinements to within 11% and the exact samples reproduce the spectral correlation function to within 2.86 standard errors on every cell. Table 5 compares generated and exact samples.

The chain outside the family. The Laplace-bond chain of Section 9.1 has the symmetric pair energy $W(a, a') = -|a' - a|/\lambda - \eta\,[V(a) + V(a')]/2$ with $V(a) = (a^2 - 1)^2/4$, bond scale $\lambda = 0.40$ and well depth $\eta = 2.0$, chosen before any fit so that its lag-one correlation and kurtosis match the base cell's; it is a reversible grid chain built from this symmetric W by the same Perron–Frobenius construction as the field's, so exact samples and an exact oracle are available from the grid. It lies outside the tilted-walk class: a Laplace bond is not a Gaussian step times a function of the arrival point.

Missing data and uncertainty. Hiding a random half or a gap of 8 or 16 of the 32 sites of a noised sequence ($t = 0.5$), and drawing 200 posterior sequences by forward-filter backward-sample on the grid with the kernels of Section 7 without refitting, the learned site energy and spline are indistinguishable from the oracle: error ratio

Table 12. Cost of one posterior mean per sequence ($L = 32$, batch 512) of grid BP at several grid sizes N_g and of six networks, on one thread (exclusive node, $t = 0.4665$, median of nine runs), with BP's speed-up over $N_g = 401$, and on eight threads (exclusive node, median over the twelve levels; BP at $N_g = 401$).

	one thread ms	speed-up ×	eight threads ms
BP, $N_g = 41$	0.042	15.9	–
BP, $N_g = 51$	0.053	12.6	–
BP, $N_g = 61$	0.063	10.6	–
BP, $N_g = 81$	0.084	8.0	–
BP, $N_g = 101$	0.107	6.2	–
BP, $N_g = 151$	0.173	3.9	–
BP, $N_g = 201$	0.249	2.7	–
BP, $N_g = 401$	0.667	1.0	0.519
MLP	0.0018	–	0.0009
window	0.023	–	0.0041
bi-GRU	0.063	–	0.041
U-Net	0.152	–	0.029
transformer	0.208	–	0.042
TCN	0.216	–	0.037

0.9999–1.0004, interval-width ratio 0.998–1.004, coverage 0.890–0.893 of a nominal 90% interval (the oracle's own is 0.89: the grid and the quantile estimate cost about 0.01). The mixture kernel behaves like the Gaussian smoother. The Gaussian smoother with the true covariance is 1.5%–4.6% worse in error on the base cell, with 7%–9% wider intervals, and within 1% in error at $\Theta = 1$: at this noise level the exact posterior buys little over a Gaussian smoother.

Clean data. Fitting on the clean sequences instead of the noised ones helps the spline and the MDN by about a factor two but not the mixture, whose risk is 0.90–0.99 of its noised risk: its gap is its wrong drift, not the noise. The one-view channel costs the spline a factor of about two to two and a half in risk at every M.

Commitment. For a noised sequence the posterior mean of the commitment of site i is $\mathbb{E}|2P(a_i > 0 \mid \mathbf{x}) - 1|$. For the Gaussian twin the posterior of a_i is $\mathcal{N}(\mu_i, v_i)$ with $\mu_i \sim \mathcal{N}(0, \Sigma_{ii} - v_i)$ over the data, so with $R_i^2 = 1 - v_i/\Sigma_{ii}$ the commitment is $\frac{2}{\pi} \arcsin R_i$ in closed form. The twin is half-committed exactly where its posterior variance has halved. The registered readings, their verdicts and the discussion of the fit of the dilute-kink law (the local slope of $\ln \xi_{\text{phys}}$ against $1/\Theta$ rises from 0.64 to 0.87 as Θ falls from 0.29 to 0.11, against the instanton value $2\sqrt{2}/3 - \Theta/2$ with a gap shrinking roughly like Θ^2) are in the experiment registry (e50).

M. Cost tables

Training. EM's E-step is linear in the number of sequences and its M-step does not depend on it, so training time is dominated by the E-step beyond a few hundred sequences. In shared-node runs, a network fit took 9 to 38 minutes at $M = 2048$ and an EM fit 8 minutes to 5 hours (indicative: the runs shared nodes), so training is not where the structured estimator wins.

Representation. Table 13 lists, for each way of compressing a BP message to D numbers (the grid, a Fourier basis on the solver's fixed domain, a Fourier basis fitted to each message, and a cumulant expansion), the number needed to reach a given KL divergence from the true message, over the messages that exact BP produces on the linear chain with Laplace noise.

Table 13. Numbers $D^\star$ needed to carry a BP message to a relative KL of 10^{-2}, 10^{-4}, 10^{-6} (median over the cells that reach it; a dash if none does) in four representations, over the 144 messages (three correlations, twelve noise levels, four sites) that exact BP produces on the linear chain with Laplace noise; the number of those cells that reach 10^{-4} and the number in which the KL is not monotone in D.

	median $D^\star$ at KL			number of cells	
	10^{-2}	10^{-4}	10^{-6}	reaching 10^{-4}	not monotone
grid	45	151	401	144	0
Fourier, fixed	101	151	201	140	92
Fourier, adapted	33	47	47	53	10
cumulant	5	5	9	24	144